ESI The Erwin Schrödinger International
Institute for Mathematical Physics
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Carleson's Theorem: Proof, Complements, Variations

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Vienna, Preprint ESI 1397 (2003) October 24, 2003

Supported by the Austrian Federal Ministry of Education, Science and Culture
Available via http://www.esi.ac.at

Carleson's Theorem: Proof, Complements, Variations

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*This work has been supported by an NSF grant. These notes are based on a series of lectures given at the Erwin Schrödinger Institute, in Vienna Austria. I an indebted to the Institute for the opportunity to present these lectures.

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1 Introduction

L. Carleson's celebrated theorem of 1965 [14] asserts the pointwise convergence of the partial Fourier sums of square integrable functions. We give a proof of this fact, in particular the proof of Lacey and Thiele [47], as it can be presented in brief self contained manner, and a number of related results can be seen by variants of the same argument. We survey some of these variants, complements to Carleson's theorem, as well as open problems.

We are concerned with the Fourier transform on the real line, given by

$$\widehat{f}(\xi) = \int e^{-ix\xi} f(x)\, dx$$

for Schwartz functions f. For such functions, it is an important elementary fact that one has Fourier inversion,

$$f(x) = \lim_{N\to\infty} \frac{1}{2\pi} \int_{-N}^{N} \widehat{f}(\xi) e^{ix\xi}\, d\xi, \qquad x \in \mathbb{R},$$

the inversion holding for all smooth compactly supported functions f. L. Carleson's theorem asserts that this holds almost everywhere, for $f \in L^2(\mathbb{R})$. For technical reasons, we find it easier to consider the equivalent one sided inversion,

$$f(x) = \lim_{N\to\infty} \frac{1}{2\pi} \int_{-\infty}^{N} \widehat{f}(\xi) e^{ix\xi}\, d\xi. \tag{1.1}$$

Schwartz functions being dense in L^2, one need only show that the set of functions for which a.e. convergence holds is closed. The standard method for doing so is to consider the maximal function below, which we refer to as the Carleson operator

$$\mathcal{C}f(x) = \sup_N \Big| \int_{-\infty}^{N} \widehat{f}(\xi) e^{ix\xi}\, d\xi \Big|, \qquad x \in \mathbb{R}. \tag{1.2}$$

There is a straight forward proposition.

1.3 Proposition. *Suppose that the Carleson operator satisfies*

$$|\{\mathcal{C}f(x) > \lambda\}| \lesssim \lambda^{-2} \|f\|_2^2, \qquad f \in L^2(\mathbb{R}),\ \lambda > 0. \tag{1.4}$$

Then, the set of functions $f \in L^2(\mathbb{R})$ for which (1.1) holds is closed and hence all of $L^2(\mathbb{R})$.

Proof. For $f \in L^2(\mathbb{R})$, we should see that

$$L_f := \limsup_{N\to\infty} \Big| f(x) - \frac{1}{2\pi} \int_{-\infty}^{N} \widehat{f}(\xi) e^{ix\xi}\, d\xi \Big| = 0 \quad \text{a.e.}$$

To do so, we show that for all $\epsilon > 0$, $|\{L_f > \epsilon\}| \lesssim \epsilon$. We take g to be a smooth compactly supported function so that $\|f - g\|_2 \le \epsilon^{3/2}$. Now Fourier inversion holds for g, whence $L_f \le \mathcal{C}(f-g) + |f-g|$. Then, by the weak type inequality, (1.4), we have

$$|\{\mathcal{C}(f-g) > \epsilon\}| \lesssim \epsilon^{-2} \|f-g\|_2^2 \lesssim \epsilon.$$

□

This is a standard proposition, which holds in a general context, and serves as one of the prime motivations for considering maximal operators. Note in particular that we are not at this moment claiming that $\mathcal{C}$ is a bounded operator on L^2. Inequality (1.4) is the so called weak L^2 bound, and we shall utilize the form of this bound in a very particular way in the proof below.

It was one of L. Carleson's great achievements to invent a method to prove this weak type estimate.

1.5 Theorem. *The estimate (1.4) holds. As a consequence, (1.1) holds for all $f \in L^2(\mathbb{R})$, for almost every $x \in \mathbb{R}$.*

Carleson's original proof [14] was extended to L^p, $1 < p < \infty$, by R. Hunt. Charles Fefferman [23] gave an alternate proof that was influential by the explicit nature of it's "time frequency" analysis. We follow the proof of M. Lacey and C. Thiele [47]. More detailed comments on the history of the proof, and related results will come later.

The proof will have three stages, the first being an appropriate decomposition of the Carleson operator. The second being an introduction of three Lemmas, which can be efficiently combined to give the proof of our Theorem. The third being a proof of the Lemmas.

We do not keep track of the value of generic absolute constants, instead using the notation $A \lesssim B$ iff $A \leq KB$ for some constant K. And $A \simeq B$ iff $A \lesssim B$ and $B \lesssim A$. The notation $\mathbf{1}_A$ denotes the indicator function of the set A. For an operator T, $\|T\|_p$ denotes the norm of T as an operator from L^p to itself.

2 Decomposition

The Fourier transform is a constant times a unitary operator on $L^2(\mathbb{R})$. In particular, we shall take the Plancherel's identity for granted.

2.1 Proposition. *For all $f, g \in L^2(\mathbb{R})$,*

$$\langle f, g \rangle = c \langle \widehat{f}, \widehat{g} \rangle$$

for appropriate constant $c = \frac{1}{2\pi}$.

The convolution of f and ψ is given by $f * \psi(x) = \int f(x-y)\psi(y)\,dy$. We shall also assume the following Lemma.

2.2 Lemma. *If a bounded linear operator* T *on* $L^2(\mathbb{R})$ *commutes with translations, then* $Tf = \psi * f$, *where* ψ *is a distribution, which is to say a linear functional on Schwartz functions. In addition, the Fourier transform of* Tf *is given by*

$$\widehat{Tf} = \widehat{\psi}\widehat{f}.$$

Let us introduce the operators associated to translation, modulation and dilation on the real line.

$$\mathrm{Tr}_y f(x) = f(x-y), \tag{2.3}$$

$$\mathrm{Mod}_\xi f(x) = e^{i\xi x} f(x), \tag{2.4}$$

$$\mathrm{Dil}^p_\lambda f(x) = \lambda^{-1/p} f(x/\lambda), \qquad 0 < p \le \infty,\ \lambda > 0. \tag{2.5}$$

Note that the dilation operator preserves L^p norm. These operators are related through the Fourier transform, by

$$\widehat{\mathrm{Tr}_y} = \mathrm{Mod}_{-y}, \qquad \widehat{\mathrm{Mod}_\xi} = \mathrm{Tr}_\xi, \qquad \widehat{\mathrm{Dil}^2_\lambda} = \mathrm{Dil}^2_{1/\lambda} \tag{2.6}$$

And we should also observe that the Carleson operator commutes with translation and dilation operators, while being invariant under modulation operators. For any $y, \xi \in \mathbb{R}$, and $\lambda > 0$,

$$\mathrm{Tr}_y \circ \mathcal{C} = \mathcal{C} \circ \mathrm{Tr}, \qquad \mathrm{Dil}^2_\lambda \circ \mathcal{C} = \mathcal{C} \circ \mathrm{Dil}^2_\lambda, \qquad \mathcal{C} \circ \mathrm{Mod}_\xi = \mathcal{C}.$$

Thus, our mode of analysis should exhibit the same invariance properties.

We have phrased the Carleson operator in terms of the operator $P_- f(x) = \int_{-\infty}^0 \widehat{f}(\xi) e^{ix\xi}\,d\xi$, which is the Fourier projection on to negative frequencies. Specifically, since multiplication of f by an exponential is associated with a translation of $\widehat{f}$, we have

$$\mathcal{C}f = \sup_N |P_-(e^{iN\cdot} f)| \tag{2.7}$$

A characterization of the operator P_- will be useful to us.

2.8 Proposition. *Up to a constant multiple, P_- is the unique bounded operator on $L^2(\mathbb{R})$ which (a) commutes with translation (b) commutes with dilations (c) has as it's kernel precisely those functions with frequency support on the positive axis.*

Proof. Let T be a bounded operator on $L^2(\mathbb{R})$ which satisfies these three properties. Condition (a) implies that T is given by convolution with respect to a distribution. Such operators are equivalently characterized in frequency variables by $\widehat{Tf} = \tau \widehat{f}$ for some bounded function τ. Condition (b) then implies that $\tau(\xi) = \tau(\xi/|\xi|)$ for all $\xi \neq 0$. A function f is in the kernel of T iff $\widehat{f}$ is supported on the zero set of τ. Thus (c) impli es that τ is identically 0 on the positive real axis, and non–zero on the negative axis. Thus, T must be a multiple of P_-. □

We move towards the tool that will permit us to decompose Carleson operator, and take advantage of some combinatorics of the time–frequency plane. We let $\mathcal{D}$ be a choice of dyadic grids on the line. Of the different choices we can make, we take the grid to be one that is preserved under dilations by powers of 2. That is

$$\mathcal{D} = \{[j2^k, (j+1)2^k) \,:\, j,k \in \mathbb{Z}\} \tag{2.9}$$

Thus, for each interval $I \in \mathcal{D}$ and $k \in \mathbb{Z}$, the interval $2^k I = \{2^k x \,:\, x \in I\}$ is also in $\mathcal{D}$.

A *tile* is a rectangle $s \in \mathcal{D} \times \mathcal{D}$ that has area one. We write a tile as $s = I \times \omega$, thinking of the first interval as a time interval and the second as frequency. The requirement of having area one is suggested by the uncertainty principle of the Fourier transform, or alternatively, our calculation of the Fourier transform of the dilation operators in (2.6). Let $\mathcal{T}$ denote the set of all tiles.

Each dyadic interval is a union of its left and right halves, which are also dyadic. For an interval ω we denote these as ω_- and ω_+ respectively. We are in the habit of associating frequency intervals with the vertical axis. So ω_- will lie below ω_+. Associate to a tile $s = I_s \times \omega_s$ the rectangles $s_\pm = I_s \times \omega_{s\pm}$. These two rectangles play complementary roles in our proof.

Fix a Schwartz function φ with $\mathbf{1}_{[-1/9,1/9]} \le \widehat{\varphi} \le \mathbf{1}_{[-1/8,1/8]}$. Define a function associated to a tile s by

$$\varphi_s = \operatorname{Mod}_{c(\omega_{s-})} \operatorname{Tr}_{c(I_s)} \operatorname{Dil}^2_{|I_s|} \varphi \tag{2.10}$$

where $c(J)$ is the center of the interval J. Notice that φ_s has Fourier transform supported on ω_{s-}, and is highly localized in time variables around the interval I_s. That is, φ_s is essentially supported in the time–frequency plane on the rectangle $I_s \times \omega_{s-}$. Notice that the set of functions $\{\varphi_s : s \in \mathcal{T}\}$ has a set of invariances with respect to translation, modulation, and dilation that mimics thos e of the Carleson operator.

It is our purpose to devise a decomposition of the projection P_- in terms of the tiles just introduced. To this end, for a choice of $\xi \in \mathbb{R}$, let

$$Q_\xi f = \sum_{s \in \mathcal{T}} \mathbf{1}_{\omega_{s+}}(\xi) \langle f, \varphi_s \rangle \varphi_s. \tag{2.11}$$

We should consider general values of ξ for the reason that the dyadic grid distinguishes certain points as being interior, or a boundary point, to an infinite chain of dyadic intervals. Let us list some relevant properties of these operators.

2.12 Proposition. *For any ξ, the operator Q_ξ is a bounded operator on L^2, with bound independent of ξ. Its kernel contains those functions with Fourier transform supported on $[\xi, \infty)$, and it is positive semidefinite. Moreover, for each integer k,*

$$Q_\xi = \operatorname{Dil}^2_{2^{-k}} Q_{\xi 2^{-k}} \operatorname{Dil}^2_{2^k} \tag{2.13}$$

$$Q_{\xi,k} \operatorname{Tr}_{2^k} = \operatorname{Tr}_{2^k} Q_{\xi,k}, \tag{2.14}$$

where $Q_{\xi,k} = \sum_{\substack{s \in \mathcal{T} \\ |I_s| \le 2^k}} \mathbf{1}_{\omega_{s+}}(\xi) \langle f, \varphi_s \rangle \varphi_s$.

Proof. Let $\{\omega(n) : n \in \mathbb{Z}\}$ be the set of dyadic intervals for which $\xi \in \omega(n)_+$, listed in increasing order, thus $\cdots \subset \omega(n) \subset \omega(n+1) \subset \cdots$. Let $\mathcal{T}(n) = \{s \in \mathcal{T} : \omega_s = \omega(n)\}$, and

$$Q_{(n)} f = \sum_{s \in \mathcal{T}(n)} \langle f, \varphi_s \rangle \varphi_s.$$

The intervals $\omega(n)_-$ are disjoint in n, and since φ_s has frequency support in ω_{s-}, it follows that the operators $Q_{(n)}$ are orthogonal in n. The boundedness of Q_ξ reduces therefore to the uniform boundedness of $Q_{(n)}$ in n.

Two operators $Q_{(n)}$ and $Q_{(n')}$ differ by composition with a modulation operator and a dilation operator that preserves L^2 norms. Thus, it suffices to consider the L^2 norm bound of a $Q_{(n)}$ with $|I_s| = 1$ for all $s \in \mathcal{T}(n)$. Using the fact that φ_s is a rapidly decreasing function, we see that

$$|\langle \varphi_s, \varphi_{s'} \rangle| \lesssim \operatorname{dist}(I_s, I_{s'})^{-4}. \tag{2.15}$$

Now that the spatial length of the tiles is one, the tiles are separated by integral distances. Since $\sum_n n^{-4} < \infty$,

$$\begin{aligned} \|Q_{(n)} f\|_2^2 &= \sum_{s \in \mathcal{T}(n)} \sum_{s' \in \mathcal{T}(n)} \langle f, \varphi_s \rangle \langle \varphi_s, \varphi_{s'} \rangle \langle \varphi_{s'}, f \rangle \\ &\lesssim \sup_{n \in \mathbb{Z}} \sum_{s \in \mathcal{T}(n)} |\langle f, \varphi_s \rangle \langle f, \varphi_{(I_s + n) \times \omega(n)} \rangle| \\ &\lesssim \sum_{s \in \mathcal{T}(n)} |\langle f, \varphi_s \rangle|^2. \end{aligned}$$

The last inequality following by Cauchy-Schwarz. The last sum is easily controlled, by simply bringing in the absolute values. Since $|\langle f, \varphi_s \rangle|^2 \lesssim \int |f|^2 |\varphi_s| \, dx$

$$\begin{aligned} \sum_{s \in \mathcal{T}(n)} |\langle f, \varphi_s \rangle|^2 &\leq \int |f|^2 \sum_{s \in \mathcal{T}(n)} |\varphi_s| \, dx \\ &\leq \|f\|_2^2 \sup_x \sum_{s \in \mathcal{T}(n)} |\varphi_s(x)| \\ &\lesssim \|f\|_2^2 \end{aligned}$$

This completes the proof of the uniform boundedness of Q_ξ.

Since all of the functions φ_s that contribute to the definition of Q_ξ have frequency support to the left of ξ, the conclusion abut the kernel of the operator is obvious. And that it is positive semidefinite, observe that

$$\langle Q_\xi f, f \rangle = \sum_{\substack{s \in \mathcal{T} \\ \xi \in \omega_{s+}}} |\langle f, \varphi_s \rangle|^2 \geq 0. \tag{2.16}$$

In particular, $\langle Q_\xi \varphi_s, \varphi_s \rangle \neq 0$ for $s \in \mathcal{T}(n)$.

To see (2.13) recall (2.6) and our specific choice of grids, (2.9). To see (2.14), observe that if $I \in \mathcal{D}$ has length at most 2^k, then $I + 2^k$ is also in $\mathcal{D}$.

□

As the lemma makes clear, $\operatorname{Mod}_{-\xi} Q_\xi \operatorname{Mod}_\xi$ serves as an approximation to P_-. A limiting procedure will recover P_- exactly. Consider

$$(2.17) \quad Q = \lim_{Y \to \infty} \int_{B(Y)} \operatorname{Dil}_{2^{-\lambda}}^2 \operatorname{Tr}_{-y} \operatorname{Mod}_{-\xi} Q_\xi \operatorname{Mod}_\xi \operatorname{Tr}_y \operatorname{Dil}_{2^\lambda}^2 \mu(d\lambda, dy, d\xi).$$

Here, $B(Y)$ is the set $[1,2] \times [0,Y] \times [0,Y]$, and μ is normalized Lebesgue measure. Notice that the dilations are given in terms of 2^λ, so that in that parameter, we are performing an average with respect to the multiplicative Haar measure on $\mathbb{R}_+$.

Apply the right hand side to a Schwartz function f. It is easy to see that as $k \to -\infty$, the terms

$$\operatorname{Mod}_\xi \operatorname{Tr}_y \operatorname{Dil}_{2^\lambda}^2 Q_{\xi,k} f$$

tend to zero uniformly in the parameters ξ, y, and λ, with a rate that depends upon f. Here, $Q_{\xi,k}$ is as in (2.14). Similarly, as $k \to \infty$, the terms

$$\operatorname{Mod}_\xi \operatorname{Tr}_y \operatorname{Dil}_{2^\lambda}^2 (Q - Q_{\xi,k}) f$$

also tend to zero uniformly. Hence, the limit is seen to exist for all Schwartz functions. By Proposition 2.12, it follows that Q is a bounded operator on L^2. That Q is translation and dilation invariant follows from (2.14) and (2.13). Its kernel contains those functions with Fourier transform supported on $[0,\infty)$. Finally, if we verify that Q is not identically zero, we can conclude that it is a multiple of P_-. But, for e.g. $f = \operatorname{Mod}_{-1/8} \varphi$, it is easy to see that

$$\langle Q_\xi \operatorname{Mod}_\xi \operatorname{Tr}_y \operatorname{Dil}_{2^\lambda}^2 f, \operatorname{Mod}_\xi \operatorname{Tr}_y \operatorname{Dil}_{2^\lambda}^2 f \rangle > 0$$

so that $Qf \neq 0$. Thus, Q is a multiple of P_-.

We can return to the Carleson operator. An important viewpoint emphasized by C. Fefferman's proof [23] is that we should linearize the supremum. That

is we consider a measurable map $N : \mathbb{R} \to \mathbb{R}$, which specifies the value of N at whic h the supremum in (1.2) occurs. Then, it suffices to bound the operator norm of the linear (not sublinear) operator

$$P_- \operatorname{Mod}_{N(x)}$$

Considering (2.17), we set

$$\mathcal{C}_N f(x) = \sum_{s \in \mathcal{T}} \mathbf{1}_{\omega_{s+}}(N(x)) \langle f, \varphi_s \rangle \varphi_s(x).$$

Our main Lemma is then

2.18 Lemma. *There is an absolute constant K so that for all measurable functions $N : \mathbb{R} \to \mathbb{R}$, we have the weak type inequality*

$$|\{\mathcal{C}_N f > \lambda\}| \lesssim \lambda^{-2} \|f\|_2^2, \qquad \lambda > 0,\ f \in L^2(\mathbb{R}). \tag{2.19}$$

By the convexity of the weak L^2 norm, this theorem immediately implies the same estimate for $P_- \operatorname{Mod}_{N(x)}$, and so proves Theorem 1.5. The proof of the lemma is obtained by combining the three estimates detailed in the next section.

2.1 Exercises

2.20. For Schwartz functions φ and ψ, set

$$Af := \sum_{n \in \mathbb{Z}} \langle f, \operatorname{Tr}_n \varphi \rangle \operatorname{Tr}_n \psi$$

$$Bf := \int_0^1 \operatorname{Tr}_{-y} A \operatorname{Tr}_y \, dy$$

Show that B is a convolution operator, that is $Bf = \Psi * f$ for some function Ψ. Find Ψ.

2.21. Show that the identity operator is, up to a constant multiple, the unique bounded operator A on L^2 which commutes with all translation and modulation operators. That is for all $y \in \mathbb{R}$ and $\xi \in \mathbb{R}$,

$$\operatorname{Tr}_y A = A \operatorname{Tr}_y, \qquad \operatorname{Mod}_\xi A = A \operatorname{Mod}_\xi .$$

For the next two problems, set

$$A_j f := \sum_{\substack{s\in\mathcal{T}\\ |I_s|=2^j}} \langle f,\varphi_s\rangle\varphi_s$$

2.22. Show that the A_j uniformly bounded operators on $L^2(\mathbb{R})$. That is,

$$\sup_j \|A_j\|_2 < \infty.$$

Is the same statement true on $L^p(\mathbb{R})$?

2.23. Show that the operator below is a non–zero multiple of the identity operator on L^2.

$$\int_0^{2^j}\int_0^{2^{-j}} \mathrm{Tr}_{-y}\,\mathrm{Mod}_{-\xi}\,A_j\,\mathrm{Mod}_{\xi}\,\mathrm{Tr}_y\;\,d\xi dy.$$

3 The Central Lemmas

Observe that the weak type estimate is implied by

$$|\langle \mathcal{C}_N f, \mathbf{1}_E\rangle| \lesssim \|f\|_2 |E|^{1/2}, \tag{3.1}$$

for all functions f and sets E of finite measure. (In fact this inequality is equivalent to the weak L^2 bound.)

By linearity of $\mathcal{C}_N$, we may assume that $\|f\|_2 = 1$. By the invariance of $\mathcal{C}_N$ under dilations by a factor of 2 (with a change of measurable $N(x)$), we can assume that $1/2 < |E| \le 1$. Set

$$\phi_s = (\mathbf{1}_{\omega_{s+}} \circ N)\varphi_s. \tag{3.2}$$

We shall show that

$$\sum_{s\in\mathcal{T}} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_E\rangle| \lesssim 1. \tag{3.3}$$

From this point forward, the function f and the set E are fixed. We use data about these two objects to organize our proof.

Also, we may consider all sums to be taken over some finite subset of tiles. Thus, there is never any question that the sums we treat are finite, and the iterative procedures we describe will all terminate.

We need some concepts to phrase the proof. There is a natural partial order on tiles. Say that $s < s'$ iff $\omega_s \supset \omega_{s'}$ and $I_s \subset I_{s'}$. Note that the time variable of s is localized to that of s', and the frequency variabl e of s is similarly localized, up to the variability allowed by the uncertainity principle. Note that two tiles are incomparable with respect to the '$<$' partial order iff the tiles, as rectangles in the time frequency plane, do not intersect. A "ma ximal tile" will one that is maximal with respect to this partial order. See figure 1.

We call a set of tiles $\mathbf{T} \subset \mathcal{S}$ a *tree* if there is a tile $I_{\mathbf{T}} \times \omega_{\mathbf{T}}$, called the *top of the tree,* such that for all $s \in \mathbf{T}$, $s < I_{\mathbf{T}} \times \omega_{\mathbf{T}}$. We note that the top is not uniquely defined. A n important point is that a tree top specifies a location in time variable for the tiles in the tree, namely inside $I_{\mathbf{T}}$, and localizes the frequency variables, identifying $\omega_{\mathbf{T}}$ as a nominal origin.

We say that $\mathcal{S}$ *has count at most* A, and write

$$\operatorname{count}(\mathcal{S}) < A$$

iff $\mathcal{S}$ is a union $\bigcup_{\mathbf{T}\in\mathcal{T}} \mathbf{T}$, where each $\mathbf{T} \in \mathcal{T}$ is a tree, and

$$\sum_{\mathbf{T}\in\mathcal{T}} |I_{\mathbf{T}}| < A.$$

Fix $\chi(x) = (1 + |x|)^{-\kappa}$, where κ is a large constant, whose exact value is unimportant to us. Define

$$\chi_I = \operatorname{Dil}^1_{|I|} \operatorname{Tr}_{c(I)} \chi, \tag{3.4}$$

$$\operatorname{dense}(s) := \sup_{s<s'} \int_{N^{-1}(\omega_{s'})} \chi_{I_{s'}}\, dx, \tag{3.5}$$

$$\operatorname{dense}(\mathcal{S}) := \sup_{s\in\mathcal{S}} \operatorname{dense}(s), \qquad \mathcal{S} \subset \mathcal{T}.$$

The first and most natural definition of a "density" of a tile, would be $|I|_s^{-1}|N^{-1}(\omega_{s+}) \cap I_s|$. But φ is supported on the whole real line, but does

decay faster than any inverse of a polynomial. We refer to this as a "Schwartz tails problem." The definition $\int_{N^{-1}(\omega_s)} \chi_{I_s}\, dx$, as it turns out, is still not adequate. That we should take the supremum over $s < s'$ only becomes evident in the proof of the "Tree Lemma" below.

The "Density Lemma" is

3.6 Lemma. *Any subset $\mathcal{S} \subset \mathcal{T}$ is a union of $\mathcal{S}_{\text{heavy}}$ and $\mathcal{S}_{\text{light}}$ for which*

$$\text{dense}(\mathcal{S}_{\text{light}}) < \tfrac{1}{2}\,\text{dense}(\mathcal{S}),$$

and the collection $\mathcal{S}_{\text{heavy}}$ satisfies

$$\text{count}(\mathcal{S}_{\text{heavy}}) \lesssim \text{dense}(\mathcal{S})^{-1}. \tag{3.7}$$

What is significant is that this relatively simple lemma admits a non-trivial variant which is intimately linked to the tree structure and orthogonality. We should refine the notion of a tree. Call a tree $\mathbf{T}$ with top $I_{\mathbf{T}} \times \omega_{\mathbf{T}}$ a $\pm$tree iff for each $s \in \mathbf{T}$, aside from the top, $I_{\mathbf{T}} \times \omega_{\mathbf{T}} \cap I_s \times \omega_{s\pm}$ is not empty. Any tree is a union of a $+$tree and a $-$tree. If $\mathbf{T}$ is a $+$tree, observe that the rectangles $\{I_s \times \omega_{s-} \,:\, s \in \mathbf{T}\}$ are disjoint. And, by the proo f of Proposition 2.12, we see that

$$\Delta(\mathbf{T})^2 := \sum_{s\in\mathbf{T}} |\langle f, \varphi_s\rangle|^2 \lesssim \|f\|_2^2.$$

This motivates the definition

$$\text{size}(\mathcal{S}) := \sup\{|I_{\mathbf{T}}|^{-1/2}\Delta(\mathbf{T}) \,:\, \mathbf{T} \subset \mathcal{S},\ \mathbf{T} \text{ is a } +\text{tree}\}. \tag{3.8}$$

The "Size Lemma" is

3.9 Lemma. *Any subset $\mathcal{S} \subset \mathcal{T}$ is a union of $\mathcal{S}_{\text{big}}$ and $\mathcal{S}_{\text{small}}$ for which*

$$\text{size}(\mathcal{S}_{\text{small}}) < \tfrac{1}{2}\,\text{size}(\mathcal{S}),$$

and the collection $\mathcal{S}_{\text{big}}$ satisfies

$$\text{count}(\mathcal{S}_{\text{big}}) \lesssim \text{size}(\mathcal{S})^{-2}. \tag{3.10}$$

Our final Lemma relates trees, density and size. It is the "Tree Lemma."

3.11 Lemma. *For any tree* $\mathbf{T}$

$$\sum_{s\in\mathbf{T}} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_E\rangle| \lesssim |I_{\mathbf{T}}|\,\mathrm{size}(\mathbf{T})\,\mathrm{dense}(\mathbf{T}). \tag{3.12}$$

The final elements of the proof are organized as follows. Certainly, $\mathrm{dense}(\mathcal{T}) < 2$ for κ sufficiently large. We take some finite subset $\mathcal{S}$ of $\mathcal{T}$, and so certainly $\mathrm{size}(\mathcal{S}) < \infty$. If $\mathrm{size}(\mathcal{S}) < 2$, we jump to the next stage of the proof. Otherwise, we iteratively apply Lemma 3.9 to obtain subcollections $\mathcal{S}_n \subset \mathcal{S}$, $n \geq 0$, for which

$$\mathrm{size}(\mathcal{S}_n) < 2^n, \quad n > 0, \tag{3.13}$$

and $\mathcal{S}_n$ satisfies

$$\mathrm{count}(\mathcal{S}_n) \lesssim 2^{-2n}. \tag{3.14}$$

We are left with a collection of tiles $\mathcal{S}' = \mathcal{S} - \bigcup_{n>0} \mathcal{S}_n$ which has both density and size at most 2.

Now, both Lemma 3.6 and Lemma 3.9 are set up for iterative application. And we should apply them so that the estimates in (3.7) and (3.10) are of the same order. (This means that we should have density about the square of the size.) As a consequence, we can achieve a decomposition of $\mathcal{S}$ into collections $\mathcal{S}_n$, $n \in \mathbb{Z}$, which satisfy (3.13), (3.14) and

$$\mathrm{dense}(\mathcal{S}_n) < \min(2, 2^{2n}). \tag{3.15}$$

Use the estimates (3.12)—(3.15). Write $\mathcal{S}_n$ as a union of trees $\mathbf{T} \in \widetilde{\mathbf{T}}_n$, this collection of trees satisfying the estimate of (3.14). We see that

$$\begin{aligned} \sum_{s\in\mathcal{S}_n} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_E\rangle| &= \sum_{\mathbf{T}\in\widetilde{\mathbf{T}}_n} \sum_{s\in\mathbf{T}} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_E\rangle| \\ &\lesssim 2^n \min(2,2^{2n}) \sum_{\mathbf{T}\in\widetilde{\mathbf{T}}_n} |I_{\mathbf{T}}| \\ &\lesssim \min(2^{-n}, 2^n). \end{aligned} \tag{3.16}$$

This is summable over $n \in \mathbb{Z}$ to an absolute constant, and so our proof (3.3) is complete, aside from the proofs of the three key lemmas.

3.1 Exercises

3.17. Show that these two conditions are equivalent.

$$\sup_{\lambda>0} \lambda^{-2}|\{f>\lambda\}| \lesssim 1,$$
$$\int_E |f| \lesssim |E|^{1/2}$$

3.18. Let A be an operator for which $\operatorname{Dil}_{2^k}^2 A = A \operatorname{Dil}_{2^k}^2$ for all $k \in \mathbb{Z}$. Suppose that there is an absolute constant K so that for all functions $f \in L^2(\mathbb{R})$ of norm one,

$$|\{Af > 1\}| \le K.$$

Show that for all $\lambda > 0$,

$$|\{Af > \lambda\}| \lesssim \lambda^{-2}\|f\|_2^2.$$

See [20].

3.19. Show that for any +tree $\mathbf{T}$,

$$\sum_{s\in\mathbf{T}} |\langle f, \varphi_s\rangle|^2 \lesssim \int |f|^2 \operatorname{Dil}_{I_{\mathbf{T}}}^{\infty} \operatorname{Tr}_{c(I_{\mathbf{T}})} \chi \, dx.$$

Conclude that

$$|I_{\mathbf{T}}|^{-1} \sum_{s\in\mathbf{T}} |\langle f, \varphi_s\rangle|^2 \lesssim \inf_{x\in\mathbf{T}} Mf|^2(x).$$

Here, M is the maximal function,

$$Mf(x) = \sup_{t>0} (2t)^{-1} \int_{-t}^{t} |f(x-y)| \, dy.$$

4 The Density Lemma

Set $\delta = \operatorname{dense}(\mathcal{S})$.

Suppose for the moment that density had the simpler definition

$$\mathsf{dense}(s) := \sup_{s<s'} \frac{|N^{-1}(\omega_s) \cap I_s|}{|I_s|}.$$

The collection $\mathcal{S}_{\text{heavy}}$ is to be a union of trees. So to select this collection, it suffices to select the tops of the trees in this set.

Select the tops of the trees, Tops as being those tiles s with $\mathsf{dense}(s)$ exceeding $\delta/2$, which are also maximal with respect to the partial order '$<$.' The tree associated to such a tile $s \in \mathsf{Tops}$ would just be all those tiles in $\mathcal{S}$ which are less than s. The tiles in Tops are pairwise incomparable with respect to the partial order '$<$,' and so are pairwise disjoint rectangles in the time–frequency plane. And so the sets $N^{-1}(\omega_s) \cap I_s \subset E$ are pairwise disjoint, and each as measure at least $\delta|I_s|/2$. Hence the estimate below is immediate.

$$\sum_{s \in \mathsf{Tops}} |I_s| \lesssim \delta^{-1} \tag{4.1}$$

The Schwartz tails problem prevents us from using this very simple estimate to prove this lemma, but in the present context, the Schwarz tails are a weak enemy at best. Let Tops be those $s \in \mathcal{S}$ which have $\text{dense}(s) > \delta/2$ and are maximal wi th respect to '$<$.' It suffices to show (4.1). For an integer $k \geq 0$, and small constant c, let $\mathcal{S}_k$ be those $s \in \mathsf{Tops}$ for which

$$|2^k I_s \cap N^{-1}(\omega_s)| \geq c 2^{2k} \delta |I_s|. \tag{4.2}$$

Every tile in Tops will be in some $\mathcal{S}_k$, with c sufficiently small, and so it suffices to show that

$$\sum_{s \in \mathcal{S}_k} |I_s| \lesssim 2^{-k} \delta^{-1}. \tag{4.3}$$

Fix k. Select from $\mathcal{S}_k$ a subset $\mathcal{S}_k'$ of tiles satisfying $\{2^k I_s \times \omega_s \,:\, s \in \mathcal{S}_k'\}$ are pairwise disjoint, and if $s \in \mathcal{S}_k$ and $s' \in \mathcal{S}_k'$ are tiles such that $2^k I_s \times \omega_s$ and $2^k I_{s'} \times \omega_{s'}$ intersect, then $|I_s| \leq |I_{s'}|$. It is clearly possible to select such a subset. And since the tiles in $\mathcal{S}_k$ are incomparable with respect to '$<$', we can use (4.2) to estimate

$$\begin{aligned} \sum_{s \in \mathcal{S}_k} |I_s| &\leq 2^{k+1} \sum_{s' \in \mathcal{S}_k'} |I_{s'}| \\ &\leq \frac{2}{c} 2^{-k} \delta^{-1}. \end{aligned}$$

That is, we see that (4.3) holds, completing our proof.

4.1 Exercises

4.4. Let $\mathcal{S}$ be a set of tiles for which there is a constant K so that for all dyadic intervals J,

$$\sum_{\substack{s\in\mathcal{S}\\ I_s\subset J}} |I_s| \lesssim K|J|.$$

Show that for all $1 \le p < \infty$, and intervals J,

$$\Big\| \sum_{\substack{s\in\mathcal{S}\\ I_s\subset J}} \mathbf{1}_{I_s} \Big\|_p \lesssim K|J|^{1/p}.$$

5 The Size Lemma

Set $\sigma = \operatorname{size}(\mathcal{S})$. We will need to construct a collection of trees $\mathbf{T} \in \widetilde{\mathbf{T}}_{\text{large}}$, with $\mathcal{S}_{\text{large}} = \bigcup_{\mathbf{T}\in\widetilde{\mathbf{T}}_{\text{large}}} \mathbf{T}$, and

$$\sum_{\mathbf{T}\in\widetilde{\mathbf{T}}_{\text{large}}} |I_{\mathbf{T}}| \lesssim \sigma^{-2}, \tag{5.1}$$

as required by (3.10).

The selection of trees $\mathbf{T} \in \widetilde{\mathbf{T}}_{\text{large}}$ will be done in conjunction with the construction of +trees $\mathbf{T}_+ \in \widetilde{\mathbf{T}}_{\text{large}+}$. This collection will play a critical role in the verification of (5.1).

The construction is recursive in nature. Initialize

$$\mathcal{S}^{\text{stock}} := \mathcal{S}, \qquad \widetilde{\mathbf{T}}_{\text{large}} := \emptyset, \qquad \widetilde{\mathbf{T}}_{\text{large}+} := \emptyset.$$

While $\operatorname{size}(\mathcal{S}^{\text{stock}}) > \sigma/2$, select a +tree $\mathbf{T}_+ \subset \mathcal{S}^{\text{stock}}$ with

$$\Delta(\mathbf{T}_+) > \frac{\sigma}{2}|I_{\mathbf{T}_+}|. \tag{5.2}$$

In addition, the top of the tree $I_{\mathbf{T}_+} \times \omega_{\mathbf{T}_+}$ should be maximal with respect to the partial order '$<$' among all trees that satisfy (5.2). And $c(\omega_{\mathbf{T}_+})$ should

be minimal, in the order of $\mathbb{R}$. Then, take $\mathbf{T}$ to be the maximal tree (without reference to sign) in $\mathcal{S}^{\text{stock}}$ with top $I_{\mathbf{T}_+} \times \omega_{\mathbf{T}_+}$.

After this tree is chosen, update

$$\mathcal{S}^{\text{stock}} := \mathcal{S}^{\text{stock}} - \mathbf{T},$$
$$\widetilde{\mathbf{T}}_{\text{large}} := \widetilde{\mathbf{T}}_{\text{large}} \cup \mathbf{T}, \qquad \widetilde{\mathbf{T}}_{\text{large}+} := \widetilde{\mathbf{T}}_{\text{large}+} \cup \mathbf{T}_+.$$

Once the while loop finishes, set $\mathcal{S}_{\text{small}} := \mathcal{S}^{\text{stock}}$ and the recursive procedure stops.

It remains to verify (5.1). This is a orthogonality statement, but one that is just weaker than true orthogonality. Note that a particular enemy is the is the situation in which $\langle \varphi_s, \varphi_{s'} \rangle \neq 0$. When $\omega_s = \omega_{s'}$, this may happen, but as we saw in the proof of Proposition 2.12, this case may be handled by direct methods. Thus we are primarily concerned with the case that e.g. $\omega_{s-} \subset_{\neq} \omega_{s'-}$.

A central part of this argument is a bit of geometry of the time–frequency plane that is encoded in the construction of the +trees above. Suppose there are two trees $\mathbf{T} \neq \mathbf{T}' \in \widetilde{\mathbf{T}}_{\text{large}+}$, and tiles $s \in \mathbf{T}$ and $s' \in \mathbf{T}'$ such that $\omega_{s-} \subset_{\neq} \omega_{s'-}$, then, it is the case that $I_{s'} \cap I_{\mathbf{T}} = \emptyset$. We refer to this property as 'strong disjointness.' It is a condition that is strictly stronger than just requiring that the sets in the time–frequency plane below are disjoint in $\mathbf{T}$.

$$\bigcup_{s \in \mathbf{T}} I_s \times \omega_{s-}, \qquad \mathbf{T} \in \widetilde{\mathbf{T}}_{\text{large}+}$$

To see that strong disjointness holds, observe that $\omega_{\mathbf{T}} \subset \omega_s \subset_{\neq} \omega_{s'-}$. Thus $\omega_{\mathbf{T}'}$ lies above $\omega_{\mathbf{T}}$. That is, in our recursive procedure, $\mathbf{T}$ was constructed first. If it were the c ase that $I_{s'} \cap I_{\mathbf{T}} \neq \emptyset$, observe that one interval would have to be contained in the other. But tiles have area one, thus, it must be the case that $I_{s'} \subset I_{\mathbf{T}}$. That means that s' would have been in the tree (the one without sign) that was removed from $\mathcal{S}^{\text{stock}}$ before $\mathbf{T}'$ was constructed. This is a contradiction which proves strong disjointness.

We use this strong disjointness condition, and the selection criteria (5.2), to prove the bound (5.1). The method of proof is closely related to the so called

TT^* method. Set $\mathcal{S}' = \bigcup_{\mathbf{T}\in\tilde{\mathbf{T}}_{\text{large}+}} \mathbf{T}$, and

$$F := \sum_{s\in\mathcal{S}''} \langle f\mathcal{S}', \varphi_s\rangle\varphi_s.$$

The operator $f \longrightarrow \langle f, \varphi_s\rangle\varphi_s$ is self–adjoint, so that

$$\begin{aligned}\sigma^2 \sum_{\mathbf{T}\in\tilde{\mathbf{T}}_{\text{large}+}} |I_{\mathbf{T}}| =& \langle f, F\rangle \\ &\leq \|f\|_2\|F\|_2\end{aligned}$$

And so, we should show that

$$\|F\|_2^2 \lesssim \sigma^2 \sum_{\mathbf{T}\in\tilde{\mathbf{T}}_{\text{large}+}} |I_{\mathbf{T}}|. \tag{5.3}$$

This will complete the proof.

This last inequality is seen by expanding the square on the left hand side. In particular, the left hand side of (5.3) is at most the sum of

$$\sum_{\substack{s,s'\in\mathcal{S}'\\ \omega_s=\omega_{s'}}} \langle f, \varphi_s\rangle\langle\varphi_s, \varphi_{s'}\rangle\langle\varphi_{s'}, f\rangle \tag{5.4}$$

$$2 \sum_{\substack{s,s'\in\mathcal{S}'\\ \omega_s\subset_{\neq}\omega_{s'}}} |\langle f, \varphi_s\rangle\langle\varphi_s, \varphi_{s'}\rangle\langle\varphi_{s'}, f\rangle| \tag{5.5}$$

For the term (5.4), we have the obvious estimate on the inner product

$$|\langle\varphi_s, \varphi_{s'}\rangle| \lesssim \Big(1 + \frac{\operatorname{dist}(I_s, I_{s'})}{|I_s|}\Big)^{-4}.$$

(Compare to (2.15).) Thus, by Cauchy–Schwarz,

$$(5.4) \lesssim \sum_{s\in\mathcal{S}'} |\langle f, \varphi_s\rangle|^2 \lesssim \sigma^2 \sum_{\mathbf{T}\in\tilde{\mathbf{T}}_{\text{large}+}} |I_{\mathbf{T}}|.$$

For the term (5.5), we need only show that for each tree $\mathbf{T}$,

$$\sum_{s\in\mathbf{T}} \sum_{\substack{s'\in\mathcal{S}'\\ \omega_s\subset_{\neq}\omega_{s'}}} |\langle f,\varphi_s\rangle\langle\varphi_s,\varphi_{s'}\rangle\langle\varphi_{s'},f\rangle|^2 \lesssim \sigma^2|I_{\mathbf{T}}|. \tag{5.6}$$

Here, $\mathcal{S}(s) := \{s' \in \mathcal{S}' - \mathbf{T} : \omega_s \subset_{\neq} \omega_{s'}\}$. The implied constant should be independent of the tree $\mathbf{T}$.

Now, the strong disjointness condition enters in two ways. For $s \in \mathbf{T}$, and $s' \in \mathcal{S}(s)$, it is the case that $I_{s'} \cap I_{\mathbf{T}} = \emptyset$. But furthermore, for $s', s'' \in \mathcal{S}(s)$, we have e.g. $\omega_{s+} \subset \omega_{s'+} \subset \omega_{s''+}$, so that $I_{s'} \cap I_{s''}$ is also empty.

At this point, we can be rather clumsy in the estimation of (5.6). The definition of size gives us the bound

$$|\langle\varphi_{s'},f\rangle| \lesssim \sqrt{|I_{s'}|}\sigma$$

And, since $\omega_s \subset \omega_{s'}$, we have $|I_s| \geq |I_{s'}|$, and I_s, and $I_{s'}$ are, in the typical situation, far apart. A trivial estimation gives

$$|\langle\varphi_s,\varphi_{s'}\rangle| \lesssim \sqrt{|I_{s'}||I_s|}\chi_{I_s}(c(I_{s'}))$$

Thus, we bound the left side of (5.6) by

$$\begin{aligned}\sum_{s\in\mathbf{T}} \sum_{s'\in\mathcal{S}(s)} |\langle f,\varphi_s\rangle\langle\varphi_s,\varphi_{s'}\rangle\langle\varphi_{s'},f\rangle| &\lesssim \sigma^2 \sum_{s\in\mathbf{T}} |I_{s'}||I_s|\chi_{I_s}(c(I_{s'}))\\ &\lesssim \sigma^2 \sum_{s\in\mathbf{T}} \int_{(I_{\mathbf{T}})^c} |I_s|\chi_{I_s}(x)\,dx\\ &\lesssim \sigma^2|I_{\mathbf{T}}|\end{aligned}$$

as is easy to verify. This completes the proof of (5.6), and so finishes the proof of Lemma 3.9.

5.1 Exercises

For the next two exercises, let $\mathbf{T}$ be a +tree and set

$$F_{\mathbf{T}} = \sum_{s\in\mathbf{T}} \langle f,\varphi_s\rangle\varphi_s$$

5.7. Show that

$$\begin{aligned} \|F_{\mathbf{T}}\|_2 &\simeq \Big[\sum_{s\in\mathbf{T}} |\langle f,\varphi_s\rangle|^2\Big]^{1/2} \\ &\lesssim \operatorname{size}(\mathbf{T})|I_{\mathbf{T}}|^{1/2}. \end{aligned}$$

5.8. Assume that $0\in\omega_{\mathbf{T}}$. Show that

$$\begin{aligned} \operatorname{size}(\mathbf{T}) &\simeq \sup_J \big[|J|^{-1}\sum_{\substack{s\in\mathbf{T}\\ I_s\subset J}} |\langle f,\varphi_s\rangle|^2\big]^{1/2} \\ &\simeq \sup_J \Big[|J|^{-1}\int_J \Big|F_{\mathbf{T}} - |J|^{-1}\int_J F_{\mathbf{T}}\Big|^2 dx\Big]^{1/2}, \end{aligned}$$

where the supremum is over all intervals J. The last quantity is the BMO norm of $F_{\mathbf{T}}$.

It is an important heuristic that for a collection $\mathcal{S}$ of pairwise incomparable tiles, the functions $\{\varphi_s : s\in\mathcal{S}\}$ are nearly orthogonal. This next exercise quantifies the sharp form of this heuristic.

5.9. Let $\mathcal{S}$ be a collection of tiles that are pairwise incomparable with respect to '$<$.' Show that for all $f\in L^2$ and all $\lambda>0$,

$$\sum_{s\in\mathcal{S}_\lambda} |I_s| \lesssim \lambda^{-2}\|f\|_2^2,$$

where $\mathcal{S}_\lambda = \{s\in\mathcal{S} : |\langle f,\varphi_s\rangle| > \lambda\sqrt{|I_s|}\}$. Note that this in an inequality about the boundedness of a sublinear operator from $L^2(\mathbb{R})$ to $L^{2,\infty}(\mathbb{R}\times\mathcal{S})$. In the latter space, one uses counting measure on $\mathcal{S}$.

Another important heuristic is that the notion of "strong disjointness" for trees is as "pairwise incomparable" is for tiles.

5.10. Let $\widetilde{\mathbf{T}}$ be a collection of strongly disjoint trees. Show that for all $f\in L^2$ and all $\lambda>0$,

$$\sum_{\mathbf{T}\in\widetilde{\mathbf{T}}_\lambda} |I_{\mathbf{T}}| \lesssim \lambda^{-2}\|f\|_2^2,$$

where $\widetilde{\mathbf{T}}_\lambda = \{\mathbf{T}\in\widetilde{\mathbf{T}} : \Delta(\mathbf{T})\geq\lambda|I_{\mathbf{T}}|\}$.

5.11. Let $\mathcal{S}$ be a collection of tiles that are pairwise incomparable with respect to '$<$.' Show that for all $2 < p < \infty$,

$$\Big[\sum_{s\in\mathcal{S}} |\langle f, \varphi_s\rangle|^p\Big]^{1/p} \lesssim \|f\|_p.$$

(Interpolate the bound in 5.9 with an endpoint estimate at L^∞.)

5.12. Let $\widetilde{\mathbf{T}}$ be a collection of strongly disjoint trees. Show that for all $2 < p < \infty$,

$$\sum_{\mathbf{T}\in\widetilde{\mathbf{T}}} \Delta(\mathbf{T})^p \lesssim \|f\|_p.$$

The L^p estimates of the previous two exercises can in some instances be improved. One such simple example is given in the next exercise.

5.13. For each integer k, show that

$$\Big\| \Big[\sum_{\substack{s\in\mathcal{T}\\ |I_s|=2^k}} \frac{|\langle f, \varphi_s\rangle|^2}{|I_s|} \mathbf{1}_{I_s}\Big]^{1/2} \Big\|_p \lesssim \|f\|_p \qquad 2 < p < \infty.$$

One way to do so is by showing that

$$\sum_{\substack{s\in\mathcal{T}\\ |I_s|=2^k}} \frac{|\langle f, \varphi_s\rangle|^2}{|I_s|} \mathbf{1}_{I_s} \lesssim (\mathrm{Dil}^1_{2^k}\, \chi) * |f|^2.$$

6 The Tree Lemma

We begin with some remarks about the maximal function, and a particular form of the same, that we shall use at a critical point of this proof. Consider the maximal function

$$Mf = \sup_{I\in\mathcal{D}} \mathbf{1}_I |\langle f, \chi_I\rangle|.$$

It is well known that this is bounded on L^2. A proof follows. Consider a linearized version of the supremum. To each $I \in \mathcal{D}$, associate a set $E(I) \subset I$,

and require that the sets $\{E(I) : I \in \mathcal{D}\}$ be pairwise disjoint. (Thus, for fixed f, $E(I)$ is that subset of I on which the supremum above is equal to $|\langle f, \chi_I\rangle|$.) Define

$$Af = \sum_{I\in\mathcal{D}} \mathbf{1}_{E(I)}\langle f, \chi_I\rangle$$

We show that $\|A\|_2$ is bounded by a constant, independent of the choice of the sets $E(I)$.

The method is that of TT^*. Note that for positive f

$$\begin{aligned} AA^*f &\le 2\sum_{I\in\mathcal{D}}\sum_{|J|\le|I|} \mathbf{1}_{E(I)}\langle\chi_I,\chi_J\rangle\langle\mathbf{1}_{E(J)}, f\rangle \\ &\lesssim \sum_{I\in\mathcal{D}} \mathbf{1}_{E(I)}\langle f,\chi_I\rangle \end{aligned}$$

It follows that

$$\begin{aligned} \|A^*\|_2^2 &= \sup_{\|f\|_2=1} \langle A^*f, A^*f\rangle \\ &= \sup_{\|f\|_2=1} \langle f, AA^*f\rangle \\ &\lesssim \|A\|_2 \end{aligned}$$

and so $\|A\|_2 \lesssim 1$, as claimed.

We shall have recourse to not only this, but a particular refinement. Let $\mathcal{J}$ be a partition of $\mathbb{R}$ into dyadic intervals. To each $J \in \mathcal{J}$, associate a subset $G(J) \subset J$, with $|G(J)| \le \delta|J|$, where $0 < \delta < 1$ is fixed. Consider

$$M_\delta f := \sum_{J\in\mathcal{J}} \mathbf{1}_{G(J)} \sup_{I\supset J} |\langle f, \chi_I\rangle| \tag{6.1}$$

Then $\|M_\delta\|_2 \lesssim \sqrt{\delta}$. The proof is

$$\begin{aligned} \int |M_\delta f|^2\, dx &= \sum_{J\in\mathcal{J}} |G(J)| \sup_{I\supset J} |\langle f,\chi_I\rangle| \\ &\le \delta \sum_{J\in\mathcal{J}} |J| \sup_{I\supset J} |\langle f,\chi_I\rangle| \\ &\le \delta \int |Mf|^2\, dx \end{aligned}$$

$$\lesssim \delta \|f\|_2^2.$$

We begin the main line of the argument. Let $\delta = \operatorname{dense}(\mathbf{T})$, and $\sigma = \operatorname{size}(\mathbf{T})$. Make a choice of signs $\varepsilon_s \in \{\pm 1\}$ such that

$$\sum_{s\in\mathbf{T}} |\langle f, \varphi_s\rangle \langle \phi_s, \mathbf{1}_E\rangle| = \int_E \sum_{s\in\mathbf{T}} \varepsilon_s \langle f, \varphi_s\rangle \phi_s \, dx.$$

By the "Schwartz tails," the integral above is supported on the whole real line. Let $\mathcal{J}$ be a partition of $\mathbb{R}$ consisting of the maximal dyadic intervals J such that $3J$ does not contain any I_s for $s \in \mathbf{T}$. It is helpful to observe that for such J if $|J| \le |I_{\mathbf{T}}|$, then $J \subset 3I_{\mathbf{T}}$. And if $|J| \ge |I_{\mathbf{T}}|$, then $\operatorname{dist}(J, I_{\mathbf{T}}) \gtrsim |J|$. The integral above is at most the sum over $J \in \mathcal{J}$ of the two terms below.

$$\sum_{\substack{s\in\mathbf{T}\\ |I_s|\le |J|}} |\langle f, \varphi_s\rangle| \int_{J\cap E} |\phi_s| \, dx \tag{6.2}$$

$$\int_{J\cap E} \Big| \sum_{\substack{s\in\mathbf{T}\\ |I_s|> |J|}} \varepsilon_s \langle f, \varphi_s\rangle \phi_s \Big| \, dx \tag{6.3}$$

Notice that for the second sum to be non–zero, we must have $J \subset 3I_{\mathbf{T}}$.

The first term (6.2) is controlled by an appeal to the "Schwartz tails." Fix an integer $n \ge 0$, and only consider those $s \in \mathbf{T}$ for which $|I_s| = 2^{-n}|J|$. Now, the distance of I_s to J is at least $\gtrsim |J|$. And,

$$|\langle f, \varphi_s\rangle| \int_{J\cap E} |\phi_s| \, dx \le \sigma\delta (|I_s|^{-1} \operatorname{dist}(I_s, J))^{-10} |I_s|.$$

The $I_s \subset I_{\mathbf{T}}$, so that summing this over $|I_s| = 2^{-n}|J|$ will give us

$$\sigma\delta 2^{-n} \min\bigl(|J|, |I_{\mathbf{T}}| (\operatorname{dist}(J, I_{\mathbf{T}})|I_{\mathbf{T}}|^{-1})^{-10}\bigr).$$

This is summed over $n \ge 0$ and $J \in \mathcal{J}$ to bound (6.2) by $\lesssim \sigma\delta |I_{\mathbf{T}}|$, as required.

Critical to the control of (6.3) is the following observation. Let

$$G(J) = J \cap \bigcup_{\substack{s\in\mathbf{T}\\ |I_s|\ge |J|}} N^{-1}(\omega_{s+}). \tag{6.4}$$

Then $|G(J)| \lesssim \delta |J|$. To see this, let J' be the next larger dyadic interval that contains J. Then $3J'$ must contain some $I_{s'}$, for $s' \in \mathbf{T}$. Let s'' be that tile with $I_{s'} \subset I_{s''}$, $|I_{s''}| = |J|$, and $\omega_{\mathbf{T}} \subset \omega_{s''}$. Then, $s' < s''$, and by the definition of density,

$$\int_{E \cap N^{-1}(\omega_{s''})} \chi_{I_{s''}} \, dx \leq \delta$$

But, for each s as in (6.4), we have $\omega_s \subset \omega_{s''}$, so that $G(J) \subset N^{-1}(\omega_{s''})$. Our claim follows.

Suppose that $\mathbf{T}$ is a $-$tree. That means that the tiles $\{I_s \times \omega_{s+} \,:\, s \in \mathbf{T}\}$ are disjoint. Then, the bound for (6.3) is no more than

$$|G(J)| \Big\| \sum_{\substack{s \in \mathbf{T} \\ |I_s| \geq |J|}} |\langle f, \varphi_s \rangle \phi_s| \Big\|_\infty \lesssim \delta \sigma |J|.$$

This is summed over $J \subset 3I_{\mathbf{T}}$ to get the desired bound.

Suppose that $\mathbf{T}$ is a $+$tree. Then, the tiles $\{I_s \times \omega_{s-} \,:\, s \in \mathbf{T}\}$ are pairwise disjoint, and we set

$$F = \mathrm{Mod}_{-c(\omega_{\mathbf{T}})} \sum_{s \in \mathbf{T}} \varepsilon_s \langle f, \varphi_s \rangle \varphi_s$$

Note that $\|F\|_2 \lesssim \sigma \sqrt{|I_{\mathbf{T}}|}$, which is a consequence of the definition of size and Proposition 2.12. Set $\tau(x) = \sup\{|I_s| \,:\, s \in \mathbf{T},\ N(x) \in \omega_{s+}\}$, and observe that for each J,

$$\sum_{\substack{s \in \mathbf{T} \\ |I_s| \geq |J|}} \varepsilon_s \langle f, \varphi_s \rangle \phi_s = \sum_{\substack{s \in \mathbf{T} \\ \tau(x) \geq |I_s| \geq |J|}} \varepsilon_s \langle f, \varphi_s \rangle \varphi_s$$

This is so since all of the intervals ω_{s+} must contain $\omega_{\mathbf{T}}$, and if $N(x) \in \omega_{s+}$, then it must also be in every other $\omega_{s'+}$ that is larger.

This last sum can be dominated by a maximal function. For any $\tau > 0$ and $J \in \mathcal{J}$, let

$$F_{\tau, J} = \mathrm{Mod}_{-c(\omega_{\mathbf{T}})} \sum_{\substack{s \in \mathbf{T} \\ \tau \geq |I_s| \geq |J|}} \varepsilon_s \langle f, \varphi_s \rangle \varphi_s$$

This function has Fourier support in the interval $[-\frac{7}{8}|J|^{-1}, -\frac{1}{4}\tau^{-1}]$. In particular, recalling how we defined φ, we can choose $\frac{1}{16} < a, b < \frac{1}{4}$ so that

$$F_{\tau,J} = (\mathrm{Dil}^1_{a|J|}\,\varphi - \mathrm{Dil}^1_{b\tau}\,\varphi) * F$$

We conclude that for $x \in J$,

$$|F_{\tau(x),J}(x)| \lesssim M_\delta F(x),$$

where M_δ is defined as in (6.1).

The conclusion of this proof is now at hand. We have

$$\begin{aligned}
\sum_{\substack{J\in\mathcal{J}\\ |J|\le 3abs\, I_{\mathbf{T}}}} \int_{G(J)} |F_{\tau(x),J}|\,dx &\lesssim \int_{\bigcup_{|J|\le 3|I_{\mathbf{T}}|} G(J)} M_\delta F\,dx \\
&\lesssim \Big|\bigcup_{|J|\le 3|I_{\mathbf{T}}|} G(J)\Big|^{1/2} \|M_\delta F\|_2 \\
&\lesssim \delta\sqrt{|I_{\mathbf{T}}|}\|F\|_2 \\
&\lesssim \sigma\delta\,|I_{\mathbf{T}}|
\end{aligned}$$

6.1 Exercises

6.5. Demonstrate the estimate below, which is somewhat cruder than the one just obtained. For all trees $\mathbf{T}$,

$$\begin{aligned}
\Big\|\sum_{s\in\mathbf{T}} \langle f,\varphi_s\rangle\phi_s\Big\|_2 &\lesssim \Big[\sum_{s\in\mathbf{T}} |\langle f,\varphi_s\rangle|^2\Big]^{1/2} \\
&\lesssim \mathrm{size}(\mathbf{T})|I_{\mathbf{T}}|^{1/2}.
\end{aligned}$$

6.6. Show that the maximal function M_δ in (6.1) admits the bounds

$$\|M_\delta\|_p \lesssim \delta^{1/p}, \qquad 1 < p < \infty.$$

This depends upon the fact that the maximal function itself maps L^p into itself, for $1 < p < \infty$.

6.7. Show that for any tree $\mathbf{T}$,

$$\Big\|\sum_{s\in\mathbf{T}}\langle g,\varphi_s\rangle\phi_s\Big\|_p \lesssim \delta^{1/p}\|g\|_p, \qquad 1<p<\infty.$$

6.8. Show that for a +tree $\mathbf{T}$,

$$\Big\|\sum_{s\in\mathbf{T}}\langle f,\varphi_s\rangle\varphi\Big\|_p \lesssim \Big\|\Big[\sum_{s\in\mathbf{T}}\frac{|\langle f,\varphi_s\rangle|^2}{|I_s|}\mathbf{1}_{I_s}\Big]^{1/2}\Big\|_p, \qquad 1<p<\infty.$$

Conclude that

$$\Big\|\sum_{s\in\mathbf{T}}\langle f,\varphi_s\rangle\varphi_s\Big\|_p \lesssim \text{ size}(\mathbf{T})|I_{\mathbf{T}}|^{1/p}, \qquad 1<p<\infty.$$

7 Carleson's Theorem on L^p, $1<p\neq 2<\infty$

We outline a proof that the Carleson maximal operator maps L^p into itself for all $1<p<\infty$. The key point is that we should obtain a distributional estimate for the model operator.

7.1 Proposition. *For $1<p<\infty$, there is an absolute constant K_p so that for all sets $E\subset\mathbb{R}$ of finite measure and measurable functions N, we have*

$$|\{|\mathcal{C}_N\mathbf{1}_E|>\lambda\}|\le K_p^p\lambda^{-p}|E|. \tag{7.2}$$

Interpolation provides the L^p inequalities. We shall in fact prove that for all sets E and F, there is a set $F'\subset F$ of measure $|F'|\ge\frac12|F|$,

$$|\langle\mathcal{C}\mathbf{1}_E,\mathbf{1}_{F'}\rangle|\lesssim \min(|E|,|F|)(1+|\log\tfrac{|E|}{|F|}|) \tag{7.3}$$

It is a routine matter to see that this estimate implies that

$$|\{|\mathcal{C}\mathbf{1}_E|>\lambda\}|\lesssim |E|\begin{cases}\lambda|\log\lambda| & \text{if } 0<\lambda<1/2\\ e^{-c\lambda} & \text{otherwise}\end{cases} \tag{7.4}$$

Here, c is an absolute constant. This distributional inequality is in fact the best that is known about the Carleson operator. See Sjölin [62] for the Walsh

case, and [63] for the Fourier case. A more recent publication proving the same point is Arias de Reyna [2]. Both authors present a proof along the lines of Carleson. We follow the weak type inequality approach of Muscalu, Tao, and Thiele [51]. The relevance of this approach to the Carleson theorem was demonstrated by Grafakos, Tao, and Terwilleger [27].

We shall find it necessary to appeal to some deeper properties of the Calderón Zygmund theory, and in particular a weak L^1 bound for the maximal function, but also the bound in (7.11) below.

In proving (7.3), we can rely upon invariance under dilations, up to a change in the measurable $N(x)$, to assume that $1/2 < |E| \le 1$. As we already know the weak L^2 estimate, (7.3) is obvious for $\frac{1}{3} < |F| < 3$. The argument then splits into two cases, that of $|F| < \frac{1}{3}$ or $|F| \ge 3$.

7.1 The case of $|F| < \frac{1}{3}$

In this case, we will take $F' = F$. Recall that $\mathcal{T}$ denotes the set of all tiles. Clearly, $\operatorname{size}(\mathcal{T}) \lesssim 1$. We repeat the argument of (3.13)—(3.16). Thus, $\mathcal{T}$ is a union of collections $\mathcal{S}_n$, for $n \ge 0$, so that

$$\operatorname{dense}(\mathcal{S}_n) \lesssim 2^{-2n}, \tag{7.5}$$

$$\operatorname{size}(\mathcal{S}_n) \lesssim \min(1, 2^{-n}|F|^{-1/2}), \tag{7.6}$$

$$\operatorname{count}(\mathcal{S}_n) \lesssim 2^{2n}|F|. \tag{7.7}$$

Then by the calculation of (3.16), we have

$$\begin{aligned} \sum_{s\in\mathcal{S}_n} |\langle \mathbf{1}_E, \varphi_s\rangle\langle \phi_s, \mathbf{1}_{F'}\rangle| &\lesssim \operatorname{dense}(\mathcal{S}_n)\operatorname{size}(\mathcal{S}_n)2^{2n}|F| \\ &\lesssim \min(|F|, |F|^{1/2}2^{-n}). \end{aligned}$$

The sum of this terms over $n \ge 0$ is no more than

$$\lesssim |F| \cdot |\log|F||$$

which is as required.

7.2 The case of $|F| \geq 3$

To define the set F', let

$$\Omega = \{M\mathbf{1}_E > C_1|F|^{-1}\}.$$

By the weak L^1 inequality for the maximal function, for an absolute choice of C_1, we have $|\Omega| < \frac{1}{2}|F|$. And we take $F' = F \cap \Omega^c$. The inner product in (7.3) is less than the sum of

$$\sum_{\substack{s\in\mathcal{T}\\ I_s\subset\Omega}} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_{F'}\rangle| \tag{7.8}$$

$$\sum_{\substack{s\in\mathcal{T}\\ I_s\not\subset\Omega}} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_{F'}\rangle| \tag{7.9}$$

These sums are handled separately.

For (7.8), let $J \subset \Omega$ be an interval such that $2^k J \subset \Omega$ but $2^{k+1}J \not\subset \Omega$. We observe two inequalities for such an interval, which are stated using the function χ_J, as defined in (3.4). The first is that

$$\int_{F'} \chi_J\,dx \leq \int_{(2^kJ)^c} \chi_J\,dx \lesssim 2^{-(\kappa-1)k}$$

Here, κ is a large constant in the definition of χ. Also, we have

$$\begin{aligned}\int_E \chi_J\,dx &\lesssim 2^k \int_E \chi_{2^{k+1}J}\,dx\\ &\lesssim 2^k \inf_{x\in 2^{k+1}J} M\mathbf{1}_E(x)\\ &\lesssim 2^k|F|^{-1}.\end{aligned}$$

The last line follows as some point in $2^{k+1}J$ must be in Ω.

Observe that among all tiles s with $I_s = J$, there is exactly one tile s with $N(x) \in \omega_{s+}$. Hence

$$\sum_{\substack{s\in\mathcal{T}\\ I_s=J}} |\langle f,\varphi_s\rangle\langle\phi_s,\mathbf{1}_{F'}\rangle| \lesssim |J| \sum_{\substack{s\in\mathcal{T}\\ I_s=J}} \langle\mathbf{1}_E,\chi_J\rangle\langle\mathbf{1}_{F'},\chi_J\rangle$$

$$\lesssim 2^{-(\kappa-2)k}|F|^{-1}|J|.$$

Recall that k is associated to how deeply J is embedded in Ω, and that Ω has measure at most $\lesssim |F|$. Hence the right hand side above can be summed over $J \subset \Omega$ to see that

$$(7.8) \lesssim 1,$$

which is better than desired.

Set $\mathcal{T}_{\text{out}} := \{s \in \mathcal{T} : I_s \not\subset \Omega\}$, which is the collection of tiles summed over in (7.9). The essential aspect of the definition of Ω is this Lemma.

7.10 Lemma.

$$\text{size}(\mathcal{T}_{\text{out}}) \lesssim |F|^{-1}.$$

Assuming the Lemma, we turn to the line of argument (3.13)—(3.16). The collection $\mathcal{T}_{\text{out}}$ can be decomposed into collections $\mathcal{S}_n$, for $n \geq 0$, for which (7.5) and (7.7) holds, and in addition

$$\text{size}(\mathcal{S}_n) \lesssim \min(|F|^{-1}, |F|^{-1/2}2^{-n}).$$

Then by the calculation of (3.16), we have

$$\sum_{s\in\mathcal{S}_n} |\langle \mathbf{1}_E, \varphi_s\rangle\langle \phi_s, \mathbf{1}_{F'}\rangle| \lesssim \min(1, |F|^{1/2}2^{-n}),$$

making the sum over $n \geq 0$ no more than $\log|F|$, as required. This completes the proof of the (7.2).

Proof. This is a consequence of the particular structure of a +tree $\mathbf{T}$, and the fact for $s \in \mathbf{T}$, the distance of $\text{supp}(\widehat{\varphi_s})$ to $\omega_{\mathbf{T}}$ is approximately $|\omega_s|$. The Calderón Zygmund theory applies, and shows that for any choice of signs $\varepsilon_s \in \{\pm 1\}$, for $s \in \mathbf{T}$,

$$(7.11) \qquad \Big|\Big\{\sum_{s\in\mathbf{T}} \varepsilon_s\langle f, \varphi_s\rangle\varphi_s > \lambda\Big\}\Big| \lesssim \lambda^{-1}\|f|I_{\mathbf{T}}|\chi_{I_{\mathbf{T}}}\|_1, \qquad \lambda > 0.$$

We apply this inequality for trees $\mathbf{T} \in \mathcal{T}_{\text{out}}$, and $f = \mathbf{1}_E$. By taking the average over all choices of signs, we can conclude a distributional estimate on the square functions

$$\Delta_{\mathbf{T}} := \Big[\sum_{s\in\mathbf{T}} \frac{|\langle \mathbf{1}_E, \varphi_s\rangle|^2}{|I_s|}\mathbf{1}_{I_s}\Big]^{1/2} \tag{7.12}$$

Namely, that for each +tree $\mathbf{T} \subset \mathcal{T}_{\text{out}}$,

$$|\{\Delta_{\mathbf{T}} > \lambda\}| \lesssim \lambda^{-1}|I_{\mathbf{T}}||F|^{-1}. \tag{7.13}$$

As this inequality applies to all subtrees of $\mathbf{T}$, it can be strengthened. (This is a reflection of the John Nirenberg inequality.) Fix the +tree $\mathbf{T} \subset \mathcal{T}_{\text{out}}$. We wish to conclude that

$$|I_{\mathbf{T}}|^{-1}\int_{I_{\mathbf{T}}} \Delta_{\mathbf{T}}^2\, dx \lesssim |F|^{-1}. \tag{7.14}$$

For a subset $\mathbf{T}' \subset \mathbf{T}$, let

$$\operatorname{sh}(\mathbf{T}') := \bigcup_{s\in\mathbf{T}'} I_s$$

be the *shadow of* $\mathbf{T}'$. Define $\Delta_{\mathbf{T}'}$ as in (7.12). And finally set

$$G(\lambda) = \sup_{\mathbf{T}'\subset\mathbf{T}} |F||\operatorname{sh}(\mathbf{T}')|^{-1}|\{\Delta_{\mathbf{T}'} > \lambda\}| \tag{7.15}$$

Notice that (7.13) implies that $G(\lambda) \lesssim \lambda^{-1}$, for $\lambda > 0$. If we show that $G(\lambda) \lesssim \lambda^{-4}$, for $\lambda > 1$, we can conclude (7.14). In fact we can show that $G(\lambda)$ decays at an exponential squared rate.

For any $\lambda > 0$, let us bound $G(\sqrt{2}\lambda)$. Fix $\mathbf{T}'$ achieving the supremum in the definition of $G(\sqrt{2}\lambda)$. Then, the set $\{\Delta_{\mathbf{T}'} > \lambda\}$ is a union of dyadic intervals. Take $\mathbf{T}'' \subset \mathbf{T}'$ to be the maximal subset whose shadow is contained in this set. Then,

$$\begin{aligned} |F|^{-1}\operatorname{sh}(\mathbf{T}')G(\sqrt{2}\lambda) &= |\{\Delta_{\mathbf{T}'} > \sqrt{2}\lambda\}| \\ &\le |\{\Delta_{\mathbf{T}''} > \lambda\}| \\ &\le |F|^{-1}\operatorname{sh}(\mathbf{T}'')G(\lambda) \\ &\le |F|^{-1}\operatorname{sh}(\mathbf{T}')G(\lambda)^2. \end{aligned}$$

We conclude that $G(\sqrt{2}\lambda) \le G(\lambda)^2$.

Fix λ_0 so that $G(\lambda_0) < \frac{1}{2}$. Induction clearly implies

$$G(2^{k/2}\lambda_0) \le 2^{-2^k}, \qquad k \ge 0, \tag{7.16}$$

which is the claimed exponential decay. Our proof of the Lemma is done. □

7.3 Exercises

7.17. Show that (7.3) implies (7.4).

7.18. Show that (7.4) implies (7.2), and obtain the estimate

$$K_p \lesssim \frac{p^2}{p-1}$$

7.19. What are the minimal conditions on the function G of (7.15) under which the exponential inequality (7.16) is true?

7.20. In (7.2), we assert the restricted weak type inequality for $1 < p < \infty$. Prove the weak type estimate for $2 < p < \infty$. That is, for $2 < p < \infty$, and $f \in L^p$ of norm one, show that

$$|\{\mathcal{C}_N f > \lambda\}| \lesssim \lambda^{-p}, \qquad \lambda > 0.$$

You may take these steps. (1) It suffices to prove the inequality above for $\lambda = 1$. (2) Define $\Omega = \{M|f|^2 > 1\}$, and show that $|\Omega| \lesssim 1$. (3) Define sums as in (7.8) and (7.9), and control each term separately. One will need to replace the Size Lemma as stated with Exercise 5.12.

8 Remarks

8.1. After L. Carleson [14] proved his theorem, R. Hunt [29] extended the argument to L^p, for $1 < p < \infty$. A similar extension, in the Walsh Paley case, was done by Billiard [9], in the case of L^2, and Sjölin [62], for all $1 < p < \infty$. The Carleson theorem has equivalent formulations on the groups $\mathbb{R}$, $\mathbb{T}$, and $\mathbb{Z}$. The last case, of the integers, was explicitly discussed by Máté [50]. This paper was overlooked until recently.

8.2. C. Fefferman [23] devised an alternate proof, which proved to be influential through it's use of methods of analysis that used both time and frequency information in an operator theoretic fashion. The proof of Lacey and Thiele [47] presented here borrows several features of that proof. The notion of tiles, and the partial order on tiles is due to C. Fefferman [23]. Likewise, the Density Lemma and the Tree Lemma, and the proof of the same, have clear antecedents in this paper.

8.3. Those familiar with the Littlewood Paley theory know that it is very useful in decoupling the scales of operators, like those of the Hilbert transform. The Carleson operator is, however, not one in which scales can be decoupled. This is another source of the interest in this Theorem.

8.4. We present the proof of the Carleson theorem on the real line due to the presence of the dilation structure.

8.5. We choose to express the Carleson operator in terms of the projection $P_{_}$. This operator is a linear combination of the identity operator and the Hilbert transform given by

$$Hf(x) := \lim_{\epsilon \to 0} \int_{\epsilon < |y| < 1/\epsilon} f(x-y)\, \frac{dy}{y}.$$

Hence, an alternate form of the Carleson operator is

$$\sup_N |H \circ \operatorname{Mod}_N f| = \sup_N \Big| \int e^{iNy} f(x-y)\, \frac{dy}{y} \Big| \tag{8.6}$$

This form is suggestive of other questions related to the Carleson Theorem, a point we rely upon below.

8.7. Despite the fact that Carleson's operator maps L^2 into itself, all three known proofs of Carleson's theorem establish the weak type bound on L^2. The strong type bound must be deduced by interpolation. On the other hand, the weak type bound is a known consequence of the pointwise convergence of Fourier series. This was observed by A. Calderón, as indicated by a footnote in [78], and is a corollary of a general observation of E.M. Stein [64].

8.8. Hunt and Young [30] have established a weighted estimate for the Carleson operator. Namely for a weight w in the class A^p, the Carleson operator maps $L^p(w)$ into itself, for $1 < p < \infty$. The method of proof utilizes the known Carleson bound, and distribution inequalities for the Hilbert transform.

8.9. The Proposition 2.8 has a well known antecedent in a characterization of (a constant times) the Hilbert transform as the unique operator A such that A is bounded operator on L^2 that commutes with dilation, is invariant under dilations, A^2 is the identity, but is not itself the identity. See [65].

8.10. The inequality (3.3) eschews all additional cancellations. It shows that all the necessary cancellation properties are already encoded in the decomposition of the operator. In addition the combinatorial model of the Carleson operator is in fact unconditionally convergent in $s \in \mathcal{T}$. This turns out to be extremely useful fact in the course of the proof: one is free to group the tiles in anyway that one likes.

8.11. The Size Lemma should be compared to Rubio de Francia's extension of the classical Littlewood Paley inequality [61]. Also see the author's recent survey of this theorem [41].

8.12. A +tree $\mathbf{T}$ is a familiar object. Aside from a modulation by $c(\omega_{\mathbf{T}})$, it shares most of the properties associated with sums of wavelets. In particular, if $0 \in \omega_{\mathbf{T}}$, note that

$$\operatorname{size}(\mathbf{T}) \simeq \Big\| \sum_{s \in \mathbf{T}} \langle f, \varphi_s \rangle \varphi_s \Big\|_{BMO}$$

where the last norm is the BMO norm.

8.13. The key instance of the Tree Lemma is that of a +tree. This case corresponds to a particular maximal function applied to a function associated to the tree. It is this point at which the supremum of Carleson's theorem is controlled by a much tamer supremum: The one in the ordinary maximal function.

8.14. The statement and proof of the size lemma Lemma 3.9 replaces the initial arguments of this type that are in Lacey and Thiele [44]. This argument has proven to be very flexible in it's application. And, in some instances it produces sharp estimates, as explained by Barrioneuvo and Lacey [8].

8.15. The set of functions $\mathcal{T}_k := \{s \in \mathcal{T} : |I_s| = 2^k\}$ is an example of a Gabor basis. For appropriate choice of φ, the operator

$$A_k f := \sum_{\substack{s \in \mathcal{T} \\ |I_s| = 2^k}} \langle f, \varphi_s \rangle \varphi_s$$

is in fact the identity operator. See the survey of I. Daubechies [22].

9 Complements and Extensions

9.1 Equivalent Formulations of Carleson's Theorem

The Fourier transform has a formulation on each of the Euclidean groups $\mathbb{R}$, $\mathbb{Z}$ and $\mathbb{T}$. Carleson's original proof worked on $\mathbb{T}$. Fefferman's proof translates very easily to $\mathbb{R}$. Máté [50] extended Carleson's proof to $\mathbb{Z}$. Each of the statements of the theorem can be stated in terms of a maximal Fourier multiplier theorem, and we have stated it as such in this paper. Inequalities for such operators can be transferred between these three Euclidean groups, and was done so by P. Auscher and M.J. Carro [7].

9.2 Fourier Series Near L^1, Part 1

The point of issue here is the sharpest determination of that integrability class which guarantees the pointwise convergence of Fourier series. The natural setting for these questions is the unit circle $\mathbb{T} = [0, 1]$, and the partial Fourier sums

$$S_N f(x) = \sum_{|n| \le N} \widehat{f}(n) e^{2\pi i n x}, \qquad \widehat{f}(n) = \int_0^1 f(x) e^{-inx}\, dx$$

In the positive direction, one seeks the "smallest" function ψ such that if $\int_{\mathbb{T}} \psi(f)\, dx < \infty$, then the Fourier series of f converge pointwise.

N. Antonov [1] has found the best result to date,

9.1 Theorem. *For all functions $f \in L(\log L)(\log\log\log L)(\mathbb{T})$, the partial Fourier series of f converges pointwise to f.*

This extends the result of P. Sjölin [62, 63], who had the result above, but with a double log where there is a triple log above. Arias de Reyna [2, 3] has noted an extension of this Theorem, in that one can define a rearrangement invariant Banach space B, so that pointwise convergence holds for all $f \in B$, and B contains $L(\log L)(\log\log\log L)$.

The method of proof takes as it's starting point, the distributional estimate of (7.4). One seeks to "extrapolate" these inequalities to the setting of the Theorem above.

9.3 Fourier Series Near L^1, Part 2

In the negative direction, A.N. Kolmogorov's fundamental example [33, 34] of an integrable function with pointwise divergent Fourier series admits a strengthening to the following statement, as obtained by Körner [37].

9.2 Theorem. *For all $\psi(x) = o(\log\log x)$, there is a function $f : [0, 2\pi] \to \mathbb{R}$ with divergent Fourier series, and $\int |f| \psi(f)\, dx < \infty$.*

The underlying method of proof was, in some essential way, unsurpassed until quite recently, when S. Konyagin [35, 36] proved

9.3 Theorem. *The previous Theorem holds assuming only*

$$\psi(x) = o\left(\sqrt{\frac{\log x}{\log\log x}}\right) \tag{9.4}$$

There is a related question, on the growth of partial sums of Fourier series of integrable functions. G.H. Hardy [28] showed that for integrable functions f, one has $S_n f = o(\log n)$ a.e., and asked if this is the best possible estimate. This question is still open, with the best result from below following from Konyagin's example. With ψ as in (9.4), there is an $f \in L^1(\mathbb{T})$ with

$$\limsup_{n\to\infty} \frac{S_n f}{\psi(n)} = \infty \qquad \text{for all } x \in \mathbb{T}.$$

Let δ_t denote the Dirac point mass at $t \in \mathbb{T}$. The method of proof is to construct measures

$$\mu = K^{-1} \sum_{k=1}^{K} \delta_{t_k}$$

a set $E \subset \mathbb{T}$ with measure at least $1/4$, and choices of integers N for which

$$\sup_{n<N} |S_n \mu(x)| \geq \psi(N).$$

Kolmogorov's example consists of uniformly distributed point masses, whereas Konyagin's example consists of point masses that have a distribution reminiscent of a Cantor set.

Kolmogorov's example has been the subject of many papers. One of the more interesting of which the author is aware of is a result of Talagrand [73]. This theorem should be compared to the results related to the Wiener Wintner Theorem, Theorem 9.7, and [38].

9.5 Theorem. *Let X_k be independent identically distributed copies of a mean zero random variable X. $X \in L(\log\log L)$ iff the series below converges to a continuous function on $\mathbb{T}$ almost surely.*

$$\sum_{k\neq 0} \frac{X_k}{k} e^{2\pi i k t}$$

9.4 The Wiener–Wintner Question

A formulation of Carleson's operator on $\mathbb{Z}$ is

$$C_{\mathbb{Z}} f(j) := \sup_{\tau} \sup_{N} \Big| \sum_{0<|k|<N} f(j-k) \frac{e^{i\tau k}}{k} \Big|.$$

See Máté [50]. Unaware of this work which followed soon after Carleson, J. Campbell and Petersen [12] considered this operator on ℓ^2, with equivalence in ℓ^p established by Assani and Petersen [5, 12]. Also see Assani, Petersen and White [6], for these and other equivalences. The latter authors had additional motivations from dynamical sy stems, which we turn to now.

A. Calderón [11] observed that inequalities for operators on $\mathbb{Z}$ which commute with translation can be transferred to discrete dynamical systems. Let (X, μ) be a probability space, and $T : X \to X$ a map which preserves μ measure.

Thus, $\mu(T^{-1}A) = \mu(A)$ for all meas urable $A \subset X$. A Carleson operator on (X, μ, T) is

$$C_{\text{mps}} f(x) := \sup_{\tau} \sup_{N} \Big| \sum_{0<|k|<N} f(T^k x) \frac{e^{i\tau k}}{k} \Big|$$

And it is a consequence of Calderón's observation and Carleson's theorem that this operator is bounded on $L^2(X)$.

There is however a curious point that distinguishes this case from the other settings of Euclidean groups. It is the case that one has pointwise convergence of

$$\lim_{N\to\infty} \sum_{0<|k|<N} f(T^k x) \frac{e^{i\tau k}}{k} \quad \text{exists for all } \tau$$

holding for almost every $x \in X$? The boundedness of the maximal function C_{mps} shows that this would hold on a closed set in $L^2(X)$. The missing ingredient is the dense class for which the convergence above holds. Unlike the setting of Euclidean groups, there is no natural dense class.

This conjecture was posed by Campbell and Petersen [12].

9.6 Conjecture. *For all measure preserving systems (X, μ, T), and all $f \in L^2(X)$, we have the following:*

$$\mu\left\{ \lim_{N\to\infty} \sum_{0<|k|<N} f(T^k x) \frac{e^{i\tau k}}{k} \quad \textit{exists for all } \tau \right\} = 1.$$

A theorem of N. Wiener and A. Wintner [77] provides a classical motivation of this question. This theorem concerns the same phenomena, but with the discrete Hilbert transform replaced by the averages.

9.7 Theorem. *For all measure preserving systems (X, μ, T), and all $f \in L^2(X)$, we have the following:*

$$\mu\left\{ \lim_{N\to\infty} N^{-1} \sum_{k=0}^{N-1} f(T^k x) e^{i\tau k} \quad \textit{exists for all } \tau \right\} = 1.$$

This theorem admits a simple proof. And note that this theorem trivially supplies a dense class in all L^p spaces, $1 \le p < \infty$.

The Wiener–Wintner theorem has several interesting variants, for which one can phrase related questions by replacing averages by Hilbert transforms. As far as is known to us, none of these questions is answered.

Here is an attractive theorem proved by E. Lesigne [48, 49].

9.8 Theorem. *For any measure preserving system (X, μ, T) and all integrable functions f, there is a subset $X_f \subset X$ of full measure so that for all $x \in X_f$, all polynomials p, and all 1–periodic functions ϕ, the limit below exists:*

$$\lim_{N\to\infty} N^{-1} \sum_{n=1}^{N} \phi(p(n)) f(T^n x)$$

Extending this theorem to the Hilbert transform would be an extraordinary accomplishment, whereas if one replaced the discrete dynamical system by flows, it could be that the corresponding result for the Hilbert transform might be within reach.

In connection to this, Arkihpov and Oskolkov [4] have proved the following theorem.

9.9 Theorem. *For all integers d,*

$$\sup_{\deg(p)=d} \Big| \sum_{n\neq 0} \frac{e^{ip(n)}}{n} \Big| < \infty.$$

with the supremum formed over all polynomials of degree d.

This is a far more subtle fact than the continuous analog stated in (9.10). Arkihpov and Oskolkov use the Hardy Littlewood Circle method of exponential sums, with the refinements of Vinogradov. See [4, 57, 58].

By Plancherel, this theorem shows that for a polynomial p which maps the integers to the integers, the operators on the integers given by

$$T_p f(j) = \sum_{n \neq 0} f(x - p(n))$$

is a bounded operator on $\ell^2(\mathbb{Z})$.

E.M. Stein and S. Wainger have established ℓ^2 mapping properties for certain Radon transforms [70, 71].

9.5 E.M. Stein's Maximal Function

A prominent theme of the research of E.M. Stein and S. Wainger concerns oscillatory integrals, with polynomial phases. It turns out to be of interest to determine what characteristics of the polynomial govern allied analytic quantities. In many instances, this characteristic is just the the degree of the polynomial. For instance, the following is a corollary to a Theorem of Stein and Wainger from 1970 [69]. Namely, that one has a bound

$$\sup_{\deg(P)=d} \Big| \int e^{ip(y)} \, \frac{dy}{y} \Big| \lesssim 1, \qquad d = 1, 2, \dots \tag{9.10}$$

A conjecture of Stein's concerns an extension of Carleson's maximal operator to one in which one forms a supremum over all polynomial choices of phase with a fixed degree. Thus,

9.11 Conjecture. *For each integer d, the maximal function below maps L^p into itself for $1 < p < \infty$.*

$$\mathcal{C}_d f(x) = \sup_{\deg(P)=d} \Big| \int e^{ip(y)} f(x - y) \, \frac{dy}{y} \Big|$$

Note that the case of $d = 1$ corresponds to Carleson's theorem. Let us set $\mathcal{C}'_d$ to be the maximal operator above, but with the the restriction that the polynomials p do not have a linear term. It is useful to make this distinction, as it is the linear terms that are intertwined with the Fourier transform.

E.M. Stein [67] considered the purely quadratic terms, and showed that $\mathcal{C}_2'$ maps L^p into itself for all $1 < p < \infty$. The essence of the matter is the bound on L^2, and there his argument is a variant on the method of TT^*, emph asizing a frequency decomposition of the operator. Lacey [40] extended this result to include the linear term and thus establish the conjecture above in the case of $d = 2$. This was done by combining the methods of Stein with those of Lacey and Thiele [47] and Thiele [74, 75].

Stein and Wainger [72] have now proved that $\mathcal{C}_d'$ is bounded on all L^p's, for all $d \geq 2$. Again the L^2 case is decisive and the argument is an application of the TT^* method, but with a spatial decomposition of the operator. It is likely that the methods of Stein and Wainger can be combined with those of Lacey and Thiele to complete the proof of this conjecture.

9.6 Fourier Series in Two Dimensions

In this section we extend the Fourier transform to functions of the plane

$$\widehat{f}(\xi) = \int f(x) e^{ix\cdot\xi}\, dx$$

where $\xi = (\xi_1, \xi_2)$, and $x = (x_1, x_2)$. The possible extensions of Carleson's theorem to the two dimensional setting are numerous. The state of our knowledge is not so great.

9.6.1 Fourier Series in Two Dimensions, Part I

Consider the pointwise convergence of the Fourier sums in the plane given by

$$\int_{tP} \widehat{f}(\xi) e^{ix\cdot\xi}\, dx$$

Here, P is polygon with finitely many sides in the plane, with the origin in it's interior. Proving the pointwise convergence of these averages is controlled by the maximal function

$$\mathcal{C}_P f := \sup_{t>0} \Big| \int_{tP} \widehat{f}(\xi) e^{ix\cdot\xi}\, dx \Big|$$

C. Fefferman [25] has observed that this maximal operator can be controlled by a sum of operators which are equivalent to the Carleson operator.

For simplicity, we just assume that the polygon is the unit square. And let

$$Qf = \int_0^\infty \int_{-\xi_1}^{\xi_1} \widehat{f}(\xi_1, \xi_2) e^{i(x_1\xi_1 + x_2\xi_2)} \, d\xi_2 d\xi_1$$

This is the Fourier projection of f onto the sector swept out by the right hand side of the square. Notice that $\mathcal{C}_P \circ Q$ is the one dimensional Carleson operator applied in the first coordinate.

Thus, $\mathcal{C}_P$ is dominated by a sum of terms which are obtained from the one dimensional Carleson operator, and so $\mathcal{C}_P$ maps L^p into itself for $1 < p < \infty$. This argument works for any polygon with a finite number of sides. While we have stressed the two dimensional aspect of this argument, it also works in any dimension.

Nevertheless, it is of some interest to consider maximal operators of the form

$$\sup_{\xi \in \mathbb{R}^2} \left| \int f(x-y) e^{i\xi \cdot y} K(y) \, dy \right|$$

where K is a Calderón Zygmund kernel. This is the question addressed by P. Sjölin [25], and more recently by Sjölin and Prestini [59] and Grafakos, Tao and Terwilleger [27].

9.6.2 Fourier Series in Two Dimensions, Part II

What other methods can be used to sum Fourier series in the plane? One method that comes to mind is over arbitrary rectangles. That is, one considers the maximal operator

$$\mathcal{R}f(x) := \sup_{\omega} \left| \int_\omega \widehat{f}(\xi) e^{i\xi \cdot x} \, d\xi \right|$$

The supremum is formed over arbitrary rectangles ω with center at the origin. C. Fefferman [24] has shown however that this is a badly behaved operator.

9.12 Proposition. *There is a bounded, compactly supported function f for which $\mathcal{R}f = \infty$ on a set of positive measure.*

This maximal operator has an alternate formulation, see (8.6), as

$$\sup_{\alpha,\beta} \Big| \iint f(x-x', y-y') e^{i(\alpha x' + \beta y')} \, \frac{dx'}{x'} \frac{dy'}{y'} \Big| \tag{9.13}$$

The example of C. Fefferman is a sum of terms $f_\lambda(x,y) = e^{i\lambda xy}\chi(x,y)$, where $\lambda > 3$, and χ is a smooth bump function satisfying e.g. $\mathbf{1}_{[-1,1]^2} \le \chi \le \mathbf{1}_{[-2,2]}$. The key observation in the construction of the example is

9.14 Lemma. *We have the estimate*

$$\mathcal{R}f_\lambda(x,y) \gtrsim \log \lambda$$

for $(x,y) \in [-\frac{1}{2}, \frac{1}{2}]^2$.

Proof. In the supremum over α and β in the definition of $\mathcal{R}$, let $\alpha = \lambda y$ and $\beta = \lambda x$, and consider

$$\begin{aligned} R(x,y) &= \int \Big[\int f_\lambda(x-x', y-y') e^{i\lambda(x'y + xy')} \, \frac{dx'}{x'} \Big] \frac{dy'}{y'} \\ &= e^{i\lambda xy} \int \Big[\int e^{i\lambda x'y'} \chi(x-x', y-y') \, \frac{dx'}{x'} \Big] \frac{dy'}{y'} \end{aligned}$$

The inside integral in the brackets admits these two estimates for all $(x,y) \in [-\frac{1}{2}, \frac{1}{2}]^2$.

$$\begin{aligned} I(x,y,y') &= \Big[\int e^{i\lambda x'y'} \chi(x-x', y-y') \, \frac{dx'}{x'} \Big] \\ &= \begin{cases} c \operatorname{sign}(\lambda y') + O(\lambda |y'|)^{-1} \\ O(1 + \lambda |y'|) \end{cases} \end{aligned}$$

c is a non–zero constant. Both of these estimates are well–known.

Then, we should estimate

$$R(x,y) = \int_{|y-y'|<1/\lambda} I(x,y,y') \frac{dy'}{y'} + \int_{|y-y'|>1/\lambda} I(x,y,y') \frac{dy'}{y'}.$$

The first term on the right is no more than $O(1)$, and the second term is $\gtrsim \log 1/\lambda$. □

This example shows that there are bounded functions f for which

$$\sup_{|\alpha|,|\beta|<N} \Big| \iint f(x-x', y-y') e^{i(\alpha x' + \beta y')} \, \frac{dx'}{x'} \frac{dy'}{y'} \Big| \gtrsim \log N.$$

It might be of interest to know if this estimate is best possible.

The integrals in (9.13) are singular integrals in the product setting. There is as of yet no positive results relating to Carleson's theorem in a product setting.

9.6.3 Fourier Series in Two Dimensions, Part III

The exponential $e^{i\xi \cdot x}$ is an eigenfunction of $-\Delta$, the positive Laplacian, with eigenvalue $|\xi|^2$. It would be appropriate to sum Fourier series according to this quantity. This concerns the operator of Fourier restriction to the unit disc

$$Tf := \int_{|\xi|<1} \widehat{f}(\xi) e^{i\xi \cdot x} \, d\xi$$

It is a famous result of C. Fefferman [26] that T is bounded on $L^2(\mathbb{R}^2)$ iff $p = 2$. The fundamental reason for this is the existence of the Besicovitch set, a set contained in a large ball, that contains a unit line segment in each dir ection, but has Lebesgue measure one. The relevance of this set is indicated by the observation that the restriction of T to very small disc placed on the boundary of the disc is well approximated by a projection onto a half space. Such a projection is a one dimensional Fourier projection performed in the normal direction to the disc. And the normal directions can point in arbitrary directions. An extension of Fefferman's argument shows that the Fourier restriction to any smooth set with a curved boundary can only be a bounded operator on L^2.

Nevertheless, the question of summability in the plane remains open. Namely,

9.15 Question. *Is it the case that the maximal operator below maps $L^2(\mathbb{R}^2)$ into weak $L^2(\mathbb{R}^2)$?*

$$\sup_{r>0} \Big| \int_{|\xi|<r} \widehat{f}(\xi) e^{i\xi \cdot x} \, d\xi \Big|$$

In the known proofs of Carleson's theorem, the truncations of singular integrals plays a distinguished role. In the proof we have presented, this is seen in the Tree Lemma, also cf. 8.13. In view of this, it interesting to suppose that if this conje cture is true, what could play a role similar to the Tree Lemma. It appears to be this.

9.16 Question. *Is it the case that the maximal operator below maps* $L^2(\mathbb{R}^2)$ *into weak* $L^2(\mathbb{R}^2)$*?*

$$\sup_{k\in\mathbb{N}} \Big| \int_{|\xi|<1+2^{-k}} \widehat{f}(\xi) e^{i\xi\cdot x}\, d\xi \Big|$$

It is conceivable that a positive answer here could lead to a proof of spherical convergence of Fourier series.

9.6.4 Fourier Series in Two Dimensions, Part IV

In order to bridge the gaps between Parts I and III, the following question comes to mind. Is there a polygon with infinitely many sides which one could sum Fourier series with respect to?

G. Mockenhoupt pointed out to the author that there is a natural first choice for P. It is a polygon P_{lac} which in the first quadrant has vertices at the points $e^{\pi i 2^{-k}}$ for $k \in \mathbb{N}$. Call this the *lacunary sided polygon.*

It is a fact due to Cordoba and R. Fefferman [21] that the lacunary sided polygon is a bounded L^p multiplier, for all $1 < p < \infty$. That is the operator below maps L^p into itself for $1 < p < \infty$.

$$T_{\text{lac}} f(x) = \int_{P_{\text{lac}}} \widehat{f}(\xi) e^{i\xi\cdot x}\, d\xi$$

This fact is in turn linked to the boundedness of the maximal function in a lacunary set of directions:

$$M_{\text{lac}} f(x) = \sup_{k\in\mathbb{N}} \sup_{t>0} (2t)^{-1} \int_{-t}^{t} |f(x - u(1, 2^{-k}))|\, du$$

Note that this is a one dimensional maximal function computed in a set of directions in the plane that, in a strong sense, is zero dimensional. Let us state as a conjecture.

9.17 Conjecture. *For $2 \le p < \infty$, the maximal function below maps $L^p(\mathbb{R}^2)$ into itself.*

$$\sup_{t>0} \Big| \int_{tP_{\text{lac}}} \widehat{f}(\xi) e^{i\xi\cdot x}\, d\xi \Big|$$

Even a restricted version of this conjecture remains quite challenging. In analogy to Question 9.16

9.18 Conjecture. *For $2 \le p < \infty$, the maximal function below maps $L^p(\mathbb{R}^2)$ into itself.*

$$\sup_{k\in\mathbb{N}} \Big| \int_{(1+2^{-k})P_{\text{lac}}} \widehat{f}(\xi) e^{i\xi\cdot x}\, d\xi \Big|$$

Another question, with a somewhat more quantitative focus, considers uniform polygons with N sides, but then seek norm bounds on these two maximal operators, on L^2 say, which grow logarithmically in N. We do not have a good conjecture as to th e correct order of growth of these constants. If one could prove that the bounds where independent of N, then the spherical summation conjecture would be a consequence.

9.7 The Bilinear Hilbert Transforms

The bilinear Hilbert transforms are given by

$$H_\alpha(f,g)(x) := \int f(x-\alpha y) g(x-y)\, \frac{dy}{y}, \qquad \alpha \in \mathbb{R},$$

with the convention that $H_0(f,g) = fHg$, and $H_\infty(f,g) = (Hf)g$. A third degenerate value is $\alpha = 1$.

These transforms commute with appropriate joint translations of f and g, and dilations of f and g. They are related to Carleson's theorem through the observation that for $\alpha \notin \{0, 1, \infty\}$, H_α enjoys an invariance property with respect to modulation. Namely,

$$H_\alpha(\mathrm{Mod}_\beta\, f, \mathrm{Mod}_{-\alpha\beta}\, g) = \mathrm{Mod}_{\beta-\alpha\beta}\, H_\alpha(f,g).$$

That is, the bilinear Hilbert transforms share the essential characteristics of the Carleson operator.

It was the study of these transforms that lead Lacey and Thiele to the proof of Carleson's theorem presented here. The bilinear Hilbert transforms are themselves interesting objects, with surprising properties. Indeed, it is natural to ask what L^p mapping properties are enjoyed by these transforms. Note that in the integral, the term dy/y is dimensionless, so that the L^p mapping properties should be those of Hölder's inequality. Thus, H_α should map $L^2 \times L^2$ into L^1. Note that this is false in the degenerate case of $\alpha = 1$, as the Hilbert transform does not preserve L^1. Neverthele ss, this was conjectured by A. Calderón in the non–degenerate cases.

See [43–46] for a proof of this theorem.

9.19 Theorem. *For $1 < p, q \leq \infty$, if $0 < 1/r = 1/p + 1/q < 3/2$, and $\alpha \notin \{0, 1, \infty\}$, then*

$$\|H_\alpha(f, g)\|_r \lesssim \|f\|_p \|g\|_q$$

We should mention that in a certain sense the proof of this theorem is easier than that for Carleson's operator. The proof outlined in [39] contains the notions of tiles, trees, and size. But the estimate that corresponds to the tree lemma is a triviality. The reason for this gain in simplicity is that there is no need for a mechanism to control a supremum.

The subject of multilinear operators with modulation invariance has inspired a large number of results, and is worthy of survey on it alone. We refer the reader to C. Thiele's article [76] for a survey of recent activity in this area.

9.8 Hilbert Transform on Smooth Families of Lines

This question has its beginnings in the Besicovitch set, which we already mentioned in connection to spherical summation of Fourier series. One may construct Besicovitch sets with these properties. For choices of $0 < \epsilon, \alpha < 1$, there is a Besic ovitch set K in the square $[0, 4]^2$ say, for which K has measure at most ϵ, and there is a function $g : K \to \mathbb{T}$, so that for a set of x's in $[0, 4]^2$

of measure $\gtrsim 1$, K contains a line segment of length one in direction $v(x)$, and v is Hölder continuous of order α.

One can ask if the Hölder continuity condition is sharp. A beautiful formulation of a conjecture in this direction is attributed to A. Zygmund.

9.20 Conjecture. *Let $v : \mathbb{R}^2 \to \mathbb{T}$ be Hölder continuous (of order 1). Then for all square integrable functions f,*

$$f(x) = \lim_{t \to 0} (2t)^{-1} \int_{-t}^{t} f(x - uv(x))\, du \qquad a.e(x)$$

This is a differentiablity question, on a choice of lines specified by v. The only stipulation is that v is Hölder continuous. This is only known under more stringent conditions on v, such as analytic due to E.M. Stein [66], or real analytic due to Bourgain [10]. There is a partial result due to N. Katz [32] (also see [31]) that demonstrates at worst "log log" blowup assuming the Hölder continuity of v. The question is open, even if one assumes that $v \in C^{1000}$.

The difficulty in this problem arises from those points at which the gradient of v is degenerate; assumptions such as analyticity certainly control such degeneracies.

E.M. Stein [66] posed the Hilbert transform variant, namely defining

$$H_v f(x) := \text{p.v.} \int_{-1}^{1} f(x - yv(x)) \frac{dy}{y},$$

is it the case that there is a constant c so that if $\|v\|_{\text{Höl}} < c$, then H_v maps $L^2(\mathbb{R}^2)$ into itself. A curious fact about this question is that this inequality, if known, implies Carleson's theorem for one dimensional Fourier series.

To see this, observe that the symbol for the transform is $\psi(\xi \cdot v(x))$, where ψ is the Fourier transform of $y^{-1}\mathbf{1}_{\{|y|<1\}}$. Suppose the vector field is of the form $v(x) = (1, \nu(x_1))$ where we need only assume that ν is Holder continuous of norm 1 say, and consider the trace of the symbol on the line $\xi_2 = -N$. Then, the symbol is $\psi((\xi_1, N) \cdot (1, \nu(x_1)) = \psi(\xi_1 - N\nu(x_1))$. We conclude that thi s symbol defines a bounded linear operator on $L^2(\mathbb{R})$, with bound

that is independent of N. That is, for *any* Lipschitz function $\nu(x_1)$, and *any* $N > 1$ the symbol $\psi(\xi_1 - N\nu(x_1))$ is the symbol of a bounded linear operator on $L^2(\mathbb{R})$. By varying N and ν, we may replace $N\nu(x_1)$ by an arbitrary measurable function. This is the substance of Carleson's theorem.

But the implication is entirely one way: A positive answer to the family of lines question seems to require techniques quite a bit more sophisticated than those that imply Carleson's theorem. Recently Lacey and Li [42] have been able to obtai n a partial answer, assuming only that the vector field has $1 + \epsilon$ derivatives.

9.21 Theorem. *Assume that $v \in C^{1+\epsilon}$ for some $\epsilon > 0$. Then the operator H_v is bounded on $L^2(\mathbb{R}^2)$. The norm of the operator is at most*

$$\|H_v\|_2 \lesssim [1 + \log_+ \|v\|_{C^{1+\epsilon}}]^{3/2}.$$

9.9 Schrödinger Operators, Scattering Transform

There is a beautiful line of investigation relating Schrödinger equations in one dimension to aspects of the Fourier transform, and in particular, Carleson's theorem. There is a further connection to scattering transforms and nonlinear Fourier analysis. All in all, these topics are extremely broad, with several different sets of motivations, and a long list of contributors.

We concentrate on a succinct way to see the connection to Carleson's theorem, an observation made explicitly by M. Christ and A. Kiselev [15, 16], also see C. Remling [60]. The basic object is a time independent Schrödinger operator on the real line,

$$H = -\frac{d^2}{dx^2} + V$$

where V is an appropriate potential on the real line. The idea is that if V is small, in some specific senses, then the spectrum of H should resemble that of $-\frac{d^2}{dx^2}$. In particular eigenfunctions should be perturbations of the exponentials.

Standard examples show that one should seek to show that for almost all λ, the eigenfunctions of energy λ, that is the solutions to

$$(H - \lambda^2 I)$$

are bounded perturbations of $e^{\pm i\lambda x}$.

Seeking such an eigenfunction, one can formally write

$$u(x) = e^{i\lambda x} + \frac{1}{i\lambda}\int_x^\infty \sin(\lambda(x-y))V(y)u(y)\,dy$$

Iterating this formula, again formally, one has

$$u(x) = e^{i\lambda x}$$

$$+ \frac{1}{i\lambda}\int_x^\infty \sin(\lambda(x-y))V(y)e^{i\lambda y}\,dy \tag{9.22}$$

$$+ \frac{1}{(i\lambda)^2}\int\int_{x\le y_1\le y_2} \sin(x-y_1)\sin(y_1-y_2)V(y_1)V(y_2)u(y_2)\,dy_1\,dy_2. \tag{9.23}$$

Observe that (9.22) no longer contains u, and is a linear combination of

$$e^{i\lambda x}\int_x^\infty e^{2i\lambda y}V(y)\,dy \tag{9.24}$$

$$e^{i\lambda x}\int_x^\infty V(y)\,dy \tag{9.25}$$

One seeks estimates of these in the mixed norm space of say, $L^2_\lambda L^\infty_x$. From such estimates, one deduces that for almost all λ, there is an eigenfunction with is a perturbation of $e^{i\lambda x}$.

Concerning (9.24), notice that if $V \in L^2$, we can, by Plancherel, regard V as $\widehat{f}$, for some $f \in L^2$. The desired estimate is then a consequence of Carleson's theorem. This is indicative of the distinguished role that L^2 plays in this subject. Also, of the intertwining of the roles of frequency and time that occur in the subject.

Concerning (9.25), unless $V \in L^1$, there is no reasonable interpretation that can be placed on this term. In practice, a different approach than the one given here must be adopted.

If one continues the expansion in (9.23), one gets a bilinear operator with features that resemble both the Carleson operator, and the bilinear Hilbert transform. See the papers by C. Muscalu, T. Tao, and C. Thiele [52,54–56].

We refer the reader to these papers by M. Christ and A. Kiselev [15–18]. For a survey of this subject, see M. Christ and A. Kiselev [19]. The reader should also consult the ongoing investigations of C. Muscalu, T. Tao, and C. Thiele [53]. This paper begins with an interesting summary of the perspective of the nonlinear Fourier transform.

10 References

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